UNIVERSITY PARTNER

UNIVERSITY OF WOLVERHAMPTON

HERALD COLLEGE KATHMANDU

Human-Computer Interaction (5CS020)

Task 3: Design & Usability Testing

White House – Real Estate Website

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I aspire to express my heartfelt gratitude to MR. Dipesh Shrestha and my tutor, MS. Gayatri Budha, for their invaluable assistance in completing this task and for sparking my interest in this module. I also appreciate the dedication of my supervisor in teaching and providing the necessary materials to accomplish this task.

This module and report have taught me the significance of the HCI module and the interface design process. The report details the steps take to design an interface, as well as the testing and evaluation of my created interface.

1. Introduction

WhiteHouse is a real estate platform where users can purchase different types of properties like homes, apartments, and plots of land. The design targets individuals ages 20 to 60, incorporating large, easy-to-read fonts and visually appealing color combination with good contrast for an improved use experience. It features an intuitive and easy-to-use interface and layout, making it straightforward to search for browse properties. Listings include detailed and clear descriptions and images, along with interactive elements. Additionally, listings and options to advertise their properties.

The interface addresses the issue of individuals of having to visit real estate agencies in person to find properties. It serves as a platform for buyers to easily search for and find properties details. Additionally, the interface enables users to contact real estate agents effortlessly and arrange property viewings. WhiteHouse includes accessibility features to assist users with disabilities, such as text-to-speech, alt text and language translation.

2. Design Process

The design process for my web interface consisted of four primary stages. The initial stage, requirements, involved activities like creating user personas, brainstorming, outlining user flows and task flows and developing mood boards. The second stage, designing, included task such as making wireframes, creating wire flows, and designing the interface. After finalizing the design, we moved on to creating and interactive prototype and conducting testing. The following paragraphs elaborate on the processes in depth:

Figure 1: Design Process

2.1 Brainstorming

Before embarking into interface design, I prioritized brainstorming meticulously noting various ideas and features for potentials inclusion. While some features made it into the design, others were left out. I began with rough sketches on paper, eventually refining them with Figma to create a polished interface.

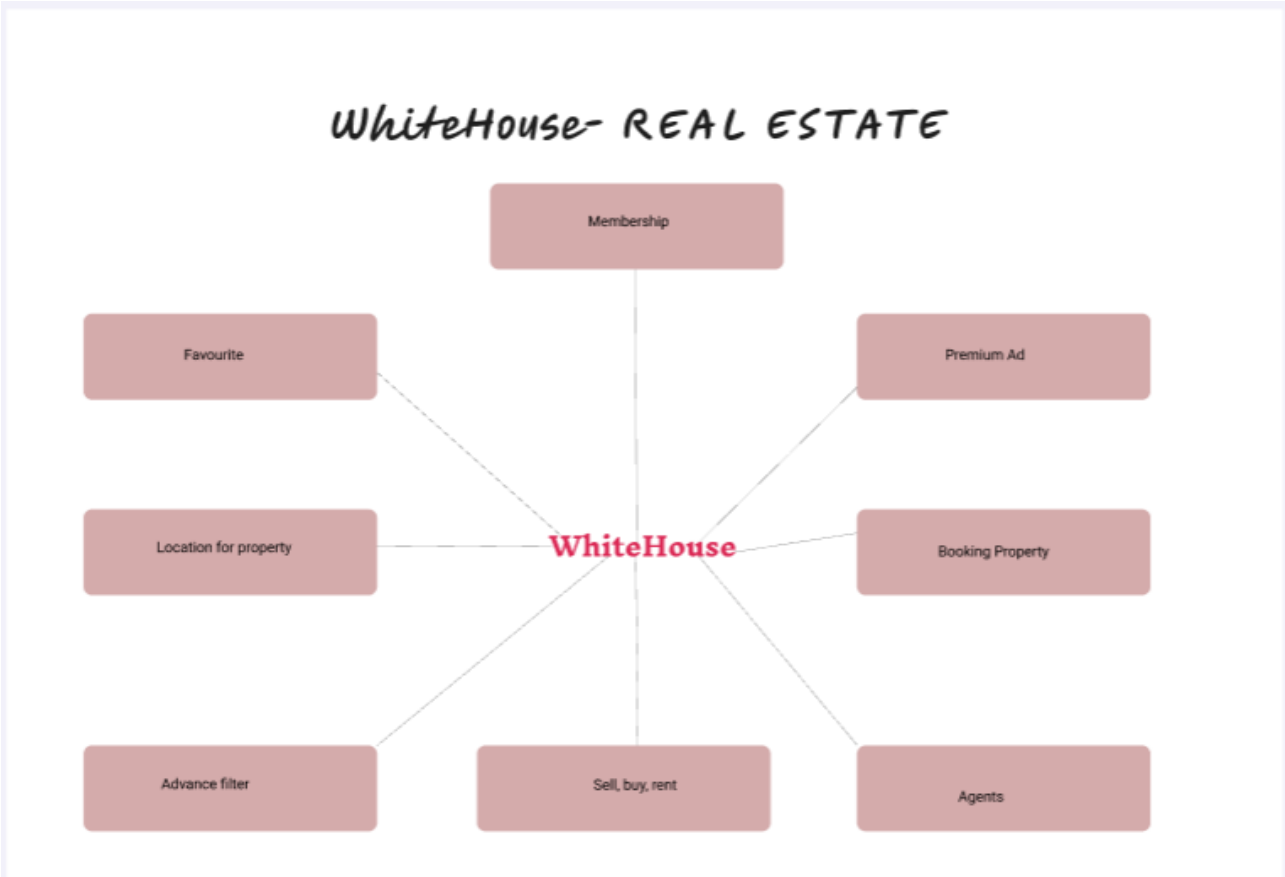


Figure 2: Brainstorming

2.2 User Persona

User persona encompass details regarding the behavior and traits of the target demographic, spanning ages 20 to 60. The three personas I developed outline users' demographics, interests, objectives, obstacles and financial inclinations. These personas serve provide thorough understanding of various requirements and desire of the audience we aim for, aiding in the development of tailored solutions and strategies.

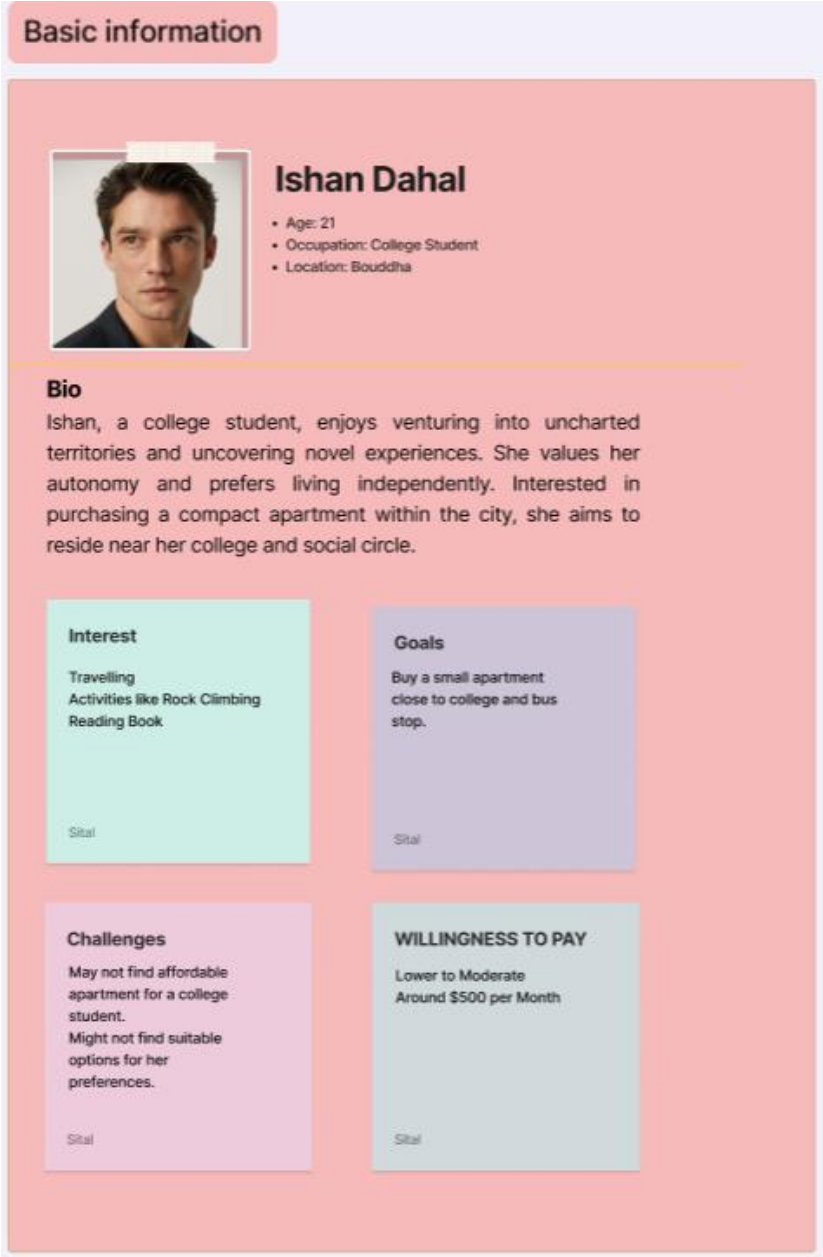


Figure 3: User personas-1

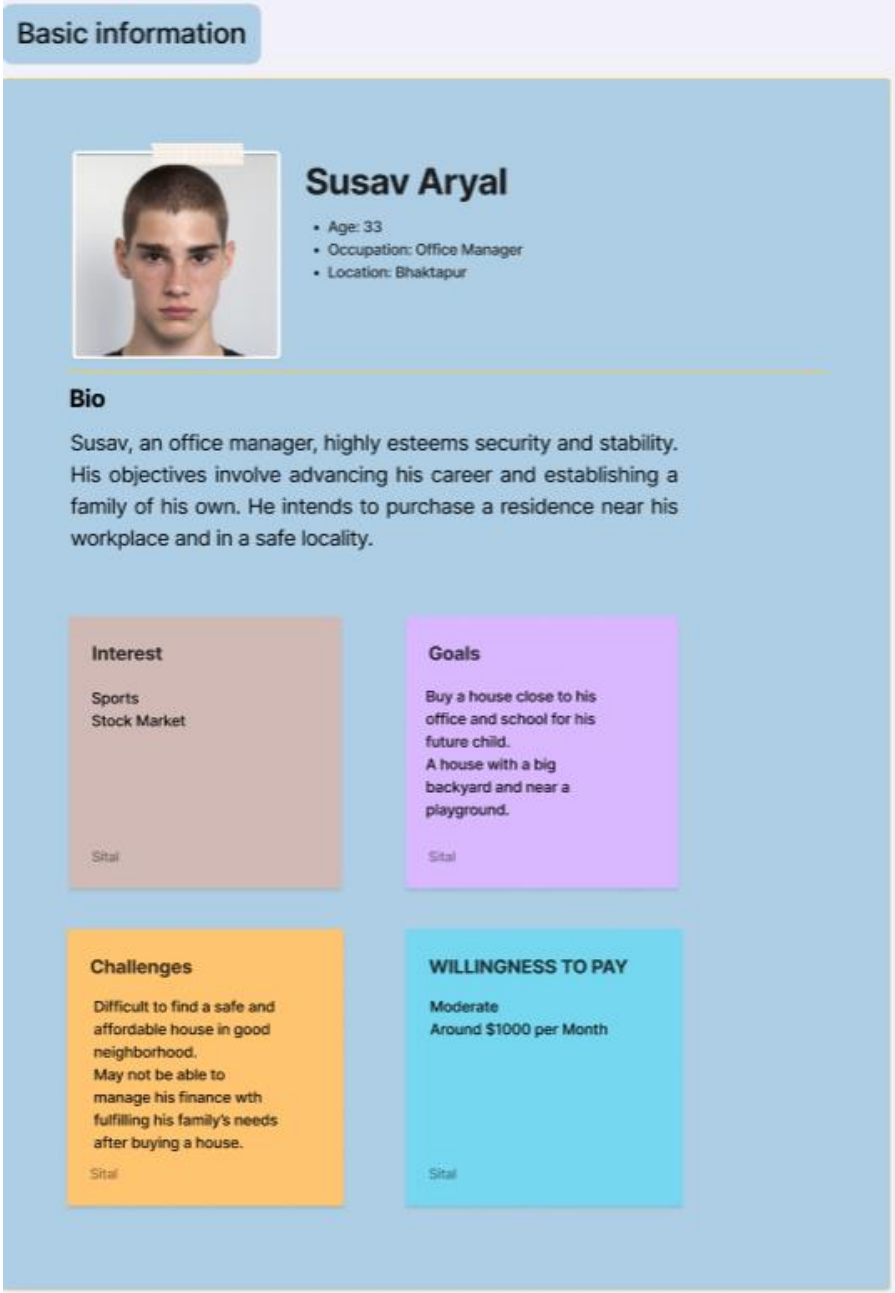


Figure 4: User personas-2

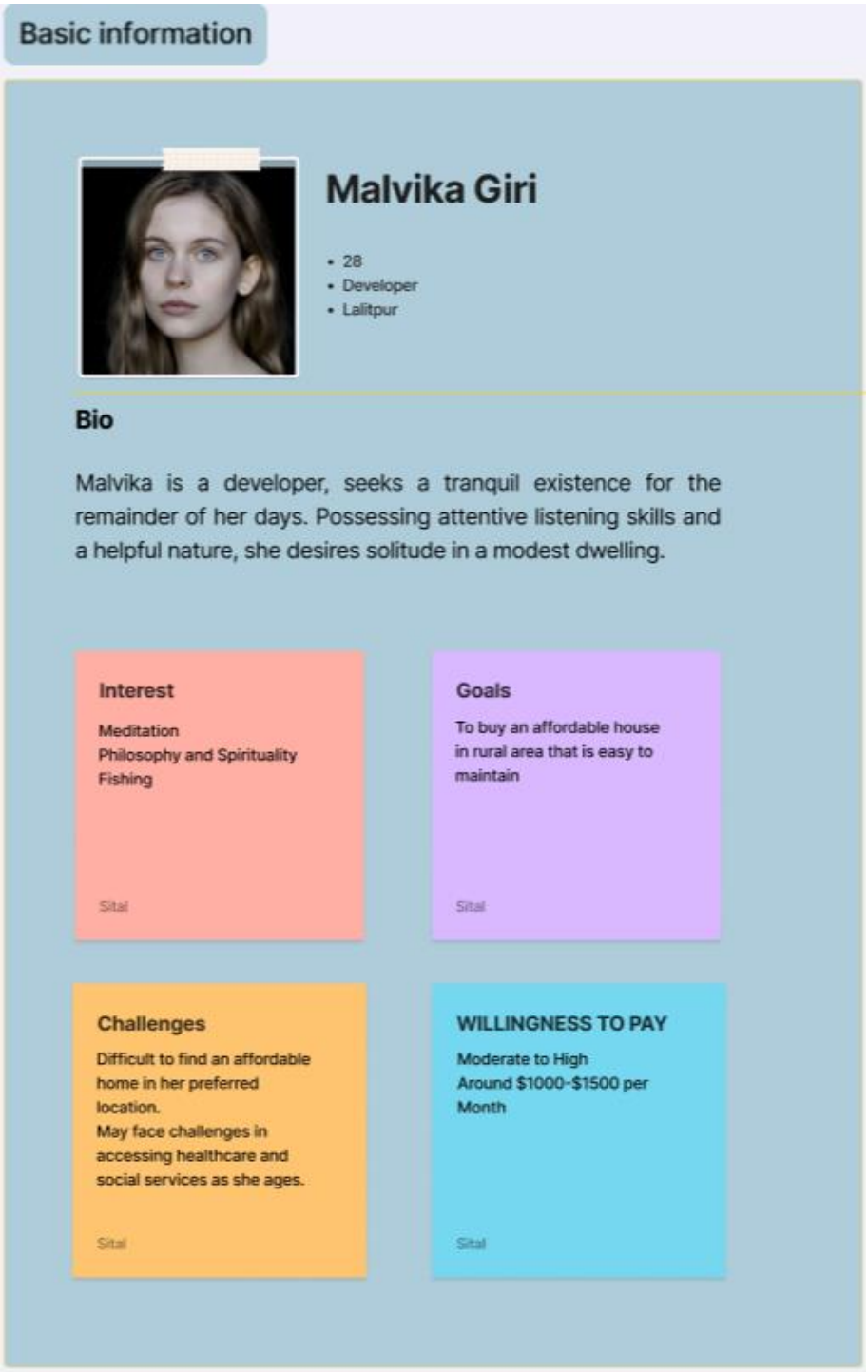


Figure 5: User personas-3

2.3 User flow

A user flow guides Users through the design, beginning with the landing page and progressing through subsequent steps to achieve their intended goal. Developing a user flow enables me to pinpoint areas for interface enhancements, ensuring a seamless user experience devoid of any usability issues. This process aids in identifying potential bottlenecks and streamlining navigation pathways ultimately enhancing overall user satisfaction and engagement.

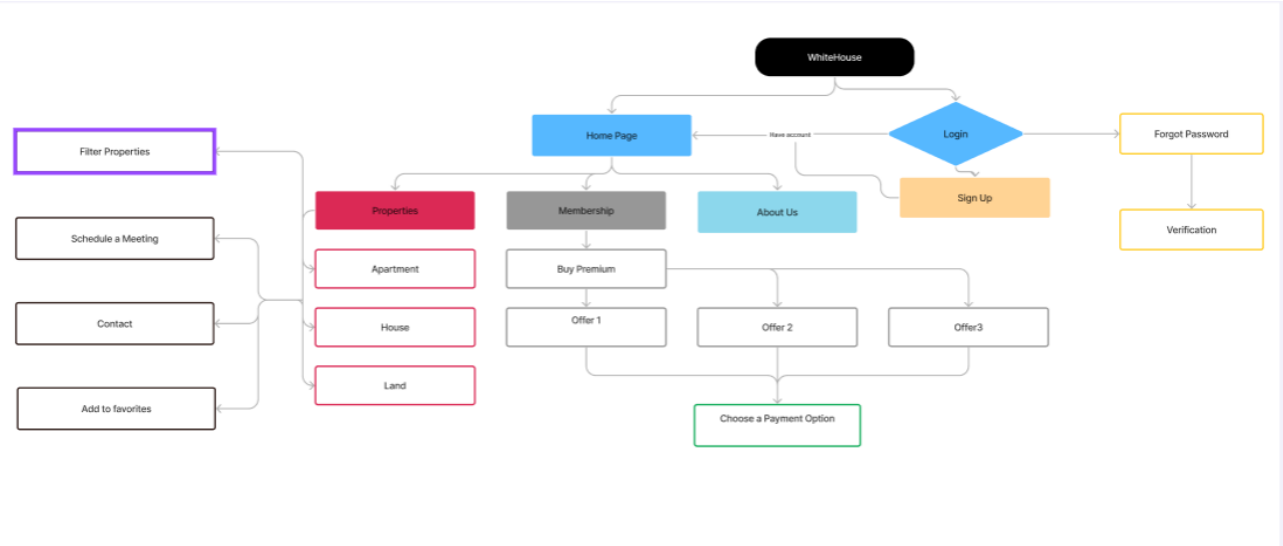


Figure 6: User flow

2.4 Mood board

A Mood board acts as a graphic compilation of diverse factors including pictures, coloration patterns, and font choices, providing a reference point for the overall design direction of a website. It helps designers maintain consistency in the view and sensation of the site by containing key component like the logo and ensuring that font sizes remain uniform across different pages. This cohesive approach ensures a harmonious visual experience for users as they navigate through the website.

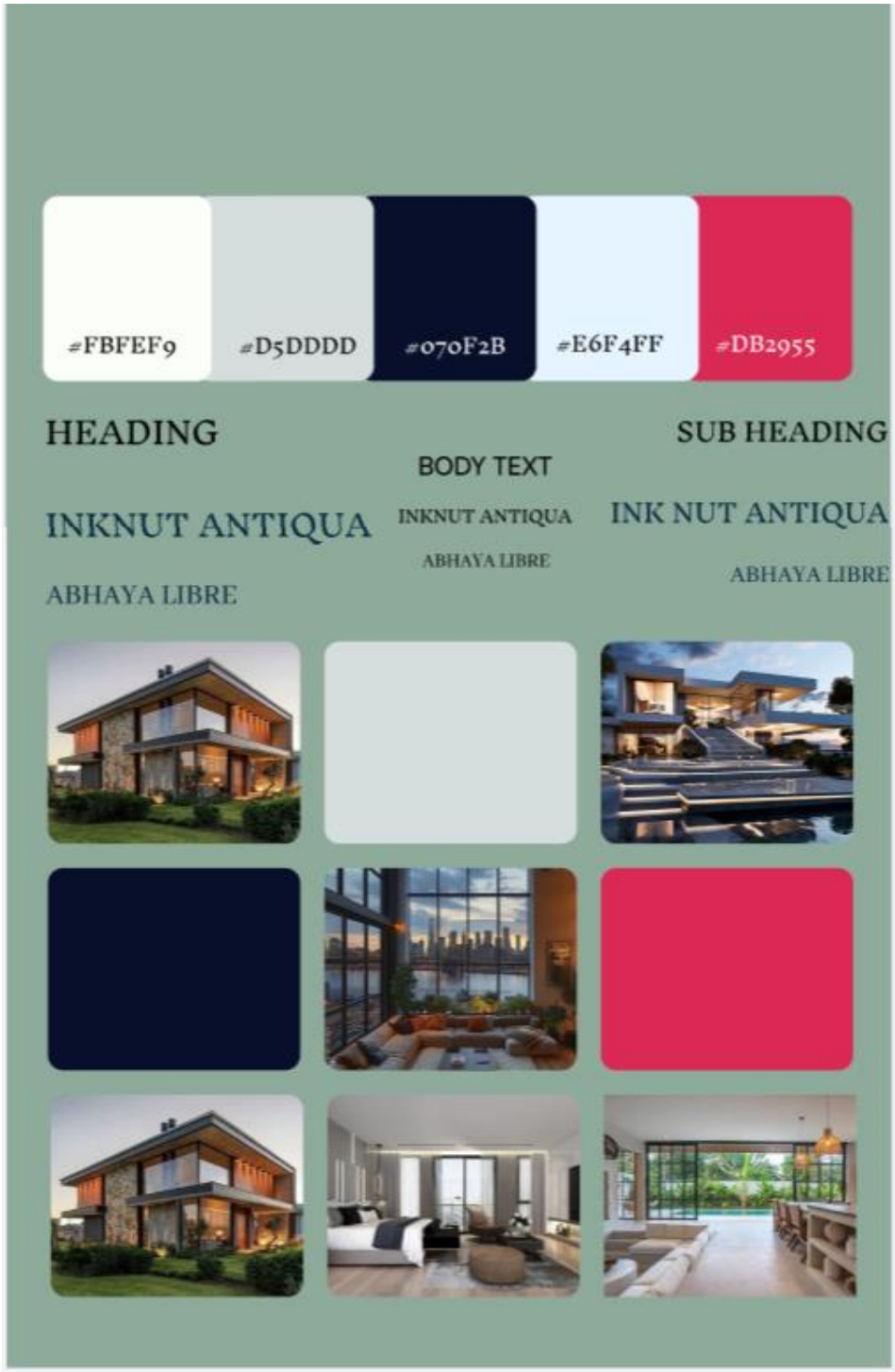


Figure 7: Mood board

2.5 Wireframe

The Initial stage in the design process involves creating a wireframe. This wireframe servers as a fundamental outline or blueprint, providing direction for designing the interface. To outline the basic design concept of my interface, I have developed key frames that include essential components.

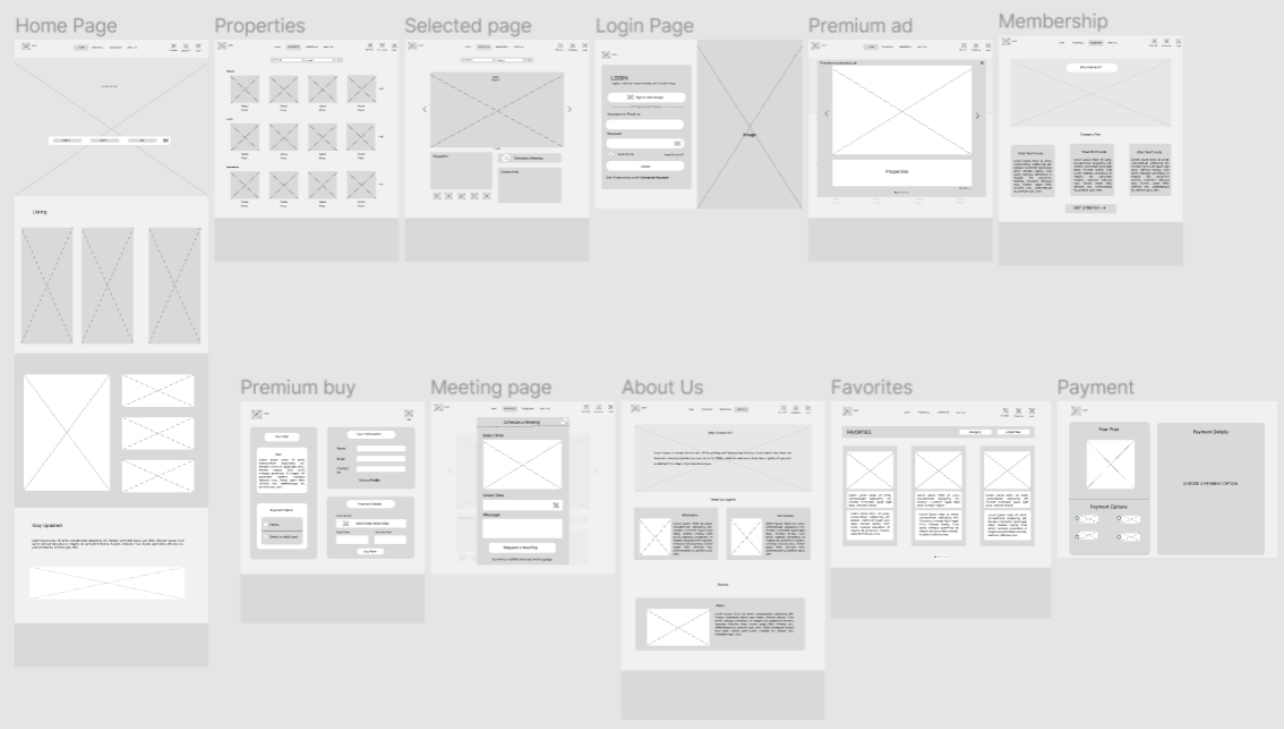


Figure 8: Wireframe

Note: Wireframe is attached in the appendix below.

2.6 Wire flow

Wire flow represents an enhanced iteration of user flow concerning the design of individual frames. It streamlines user comprehension regarding how each component within a frame connects to another. I delineated the procedure illustrating how a user can effortlessly navigate through the interface. This encompasses primary actions like browsing properties, subscribing for membership, and arranging meetings.

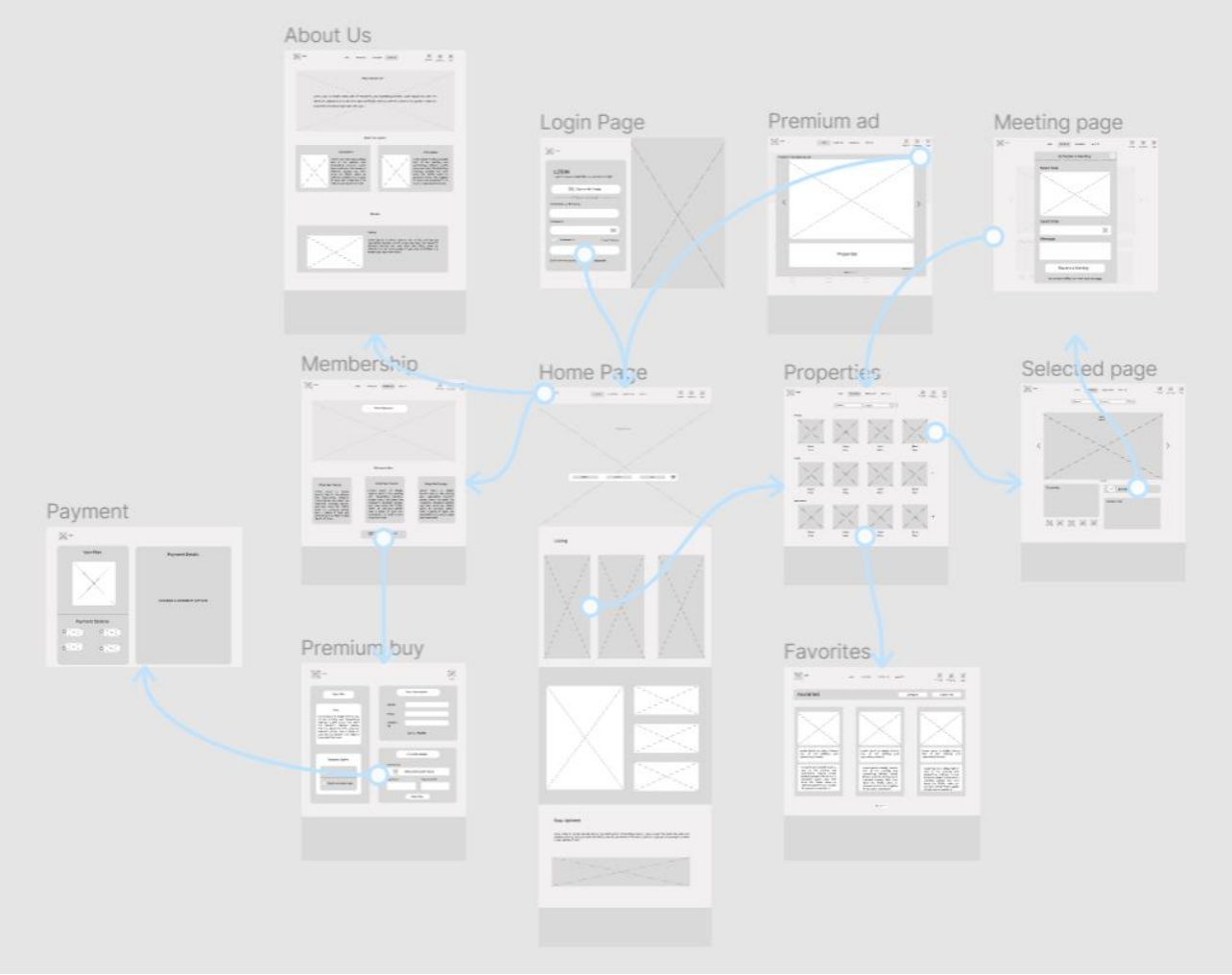


Figure 9: Wire flow

2.7 Design

Using the wire frame as a foundation, the interface design incorporates distinct color schemes, typography choices, and additional functionalities outlined in the mood board. Inspiration for the design was drawn from a variety of websites including Dribble and Pinterest.

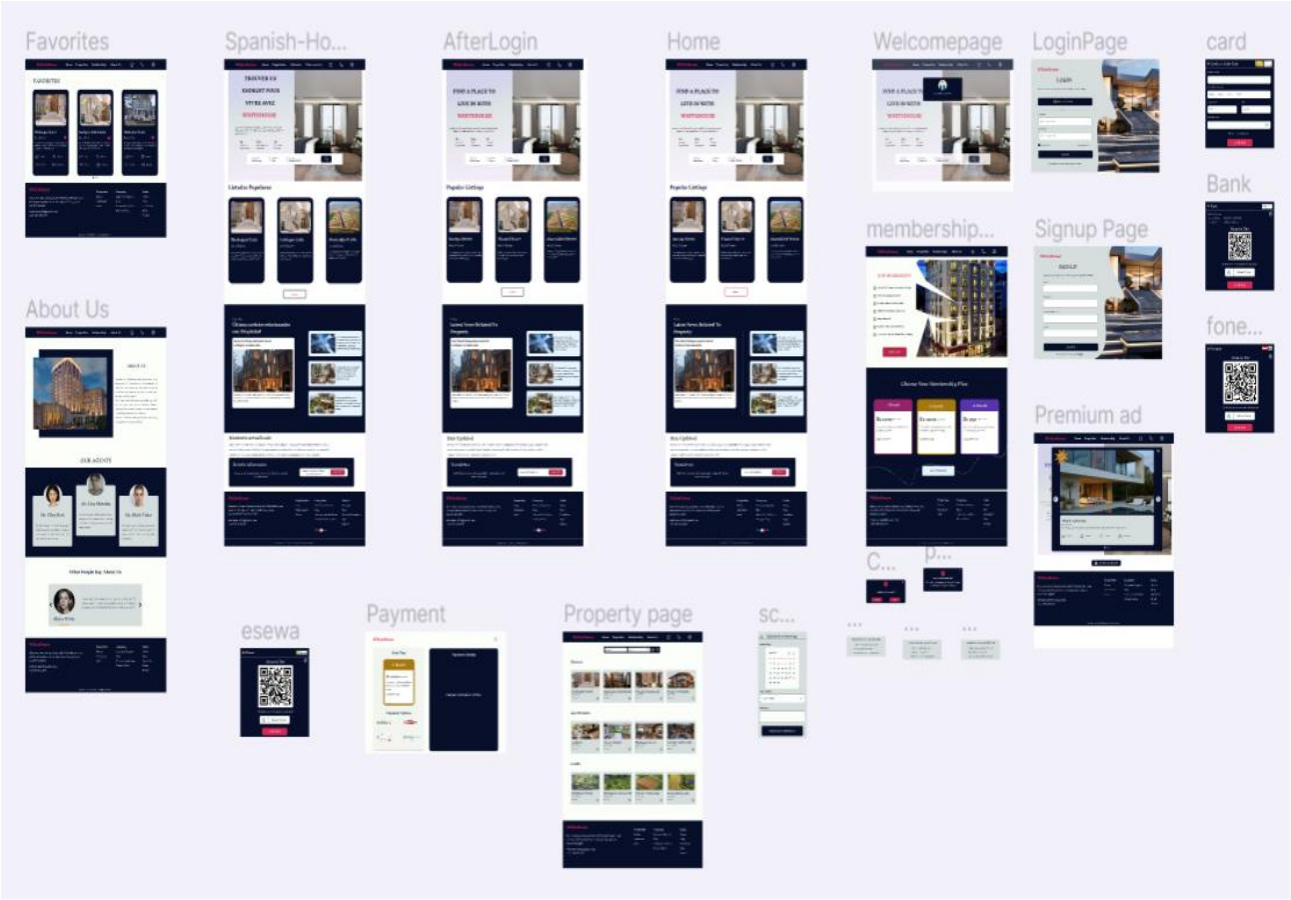


Figure 10: Design

Note: The design of key frames is attached in the appendix below.

2.8 Prototype

Following the design phase, the subsequent step is prototyping. A prototype is construction to assess and examine principles and elements, collecting user feedback for enhancement. Within an interactive prototype, buttons and navigation bar are generated to facilitate movement across various pages. This offers a visual depiction of the intended appearance and functionality of the website's frontend.

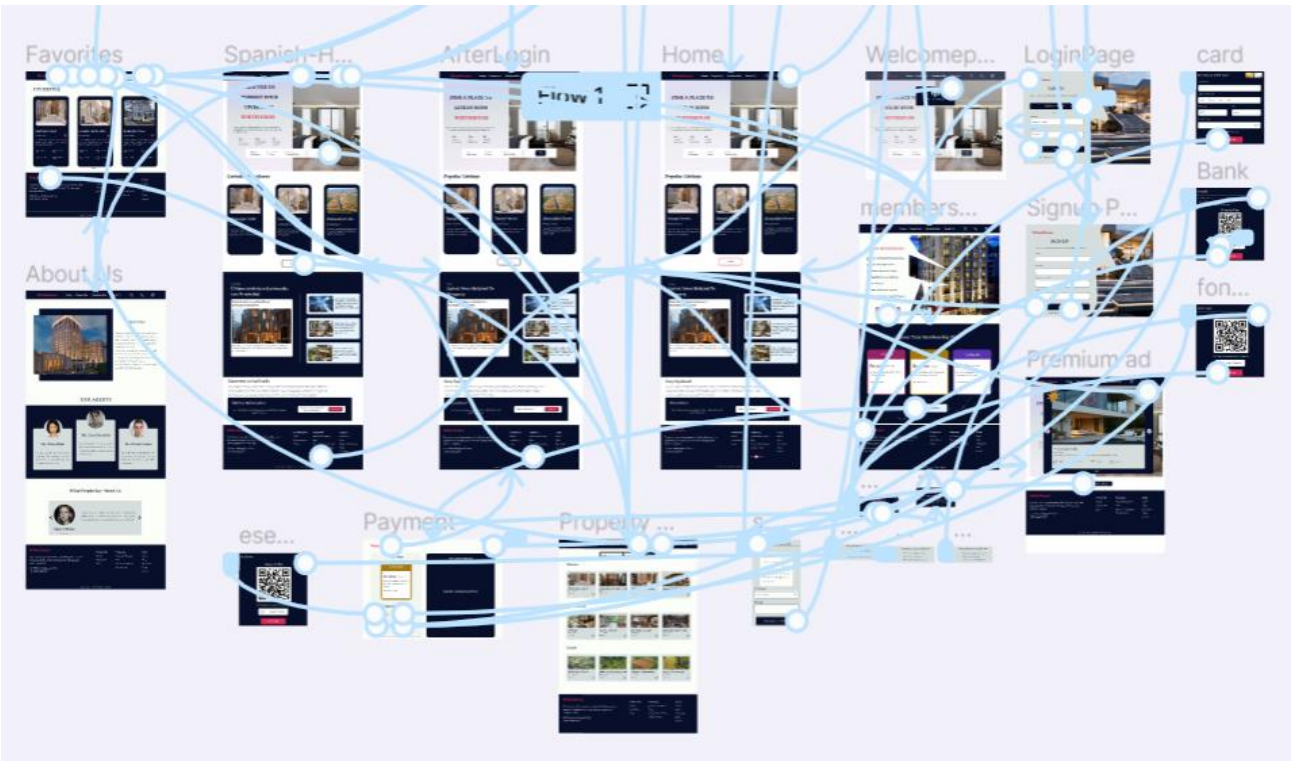


Figure 11: Prototype

3. Evaluation and Testing

Conducting an assessment and testing of my website aimed to pinpoint any issues that could impede user experience and areas requiring enhancement. The website evaluation was conducted utilizing Nielsen's Usability Heuristics and Schneiderman's 8 Golden Rule.

3.1 Usability Testing

- **Consistency and Standards**

  The website upholds uniformity across all its pages with persistent layout and navigation bar.

  The website sustains uniformity across every single page with persistent layout and navigation bar.

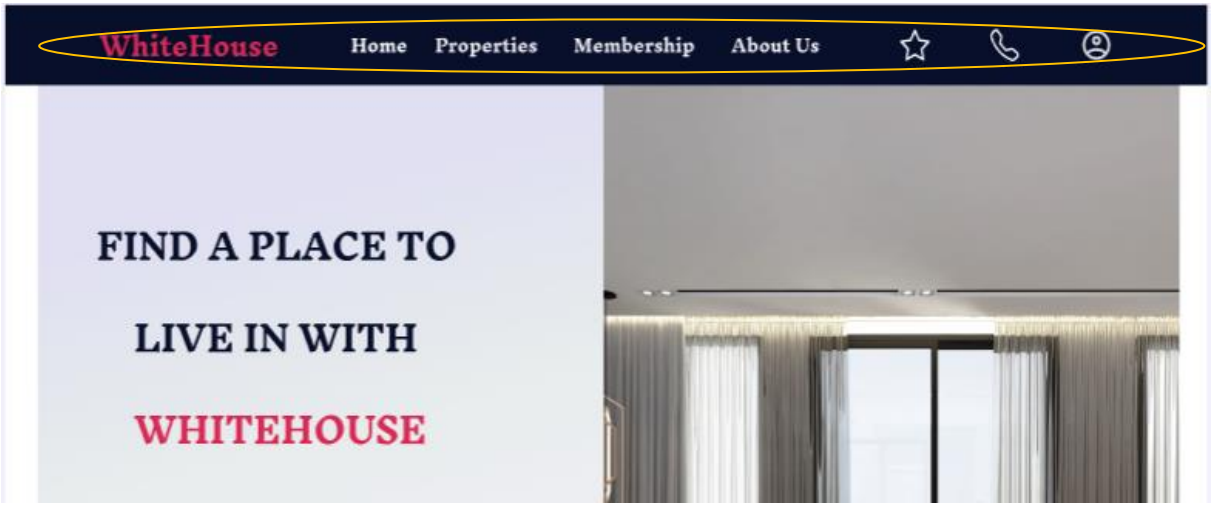


Figure 12: Consistency and standards

- **User Control and Freedom**

  WhiteHouse Enables individuals to include or exclude houses from favorites.

Figure 13: User control and freedom

3.2 User Testing

The website underwent testing by two users who assessed its usability and accessibility. OBS studio was used to create a video for the user testing process. In the video, participants initially attempted to navigate the website without prior knowledge of its functionality. Subsequently, they completed a form containing questions aimed at evaluating the website.

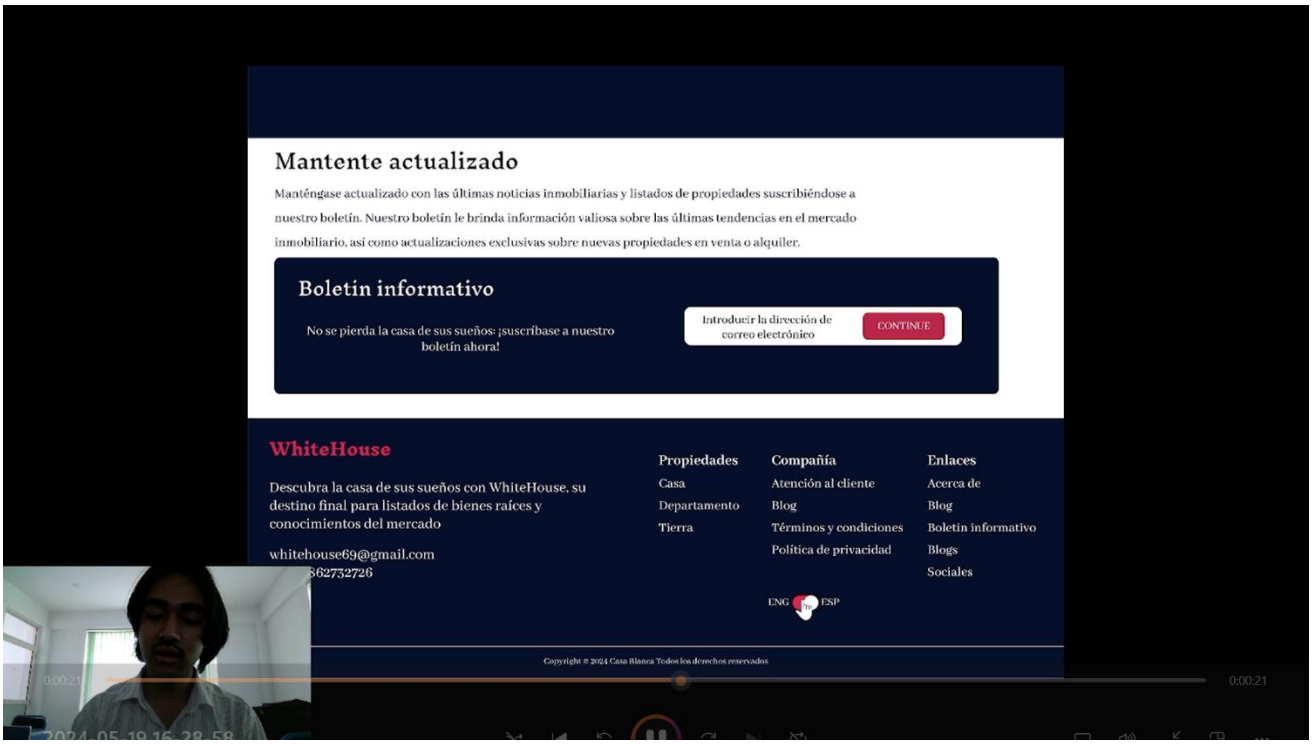


Figure 14: User testing-1

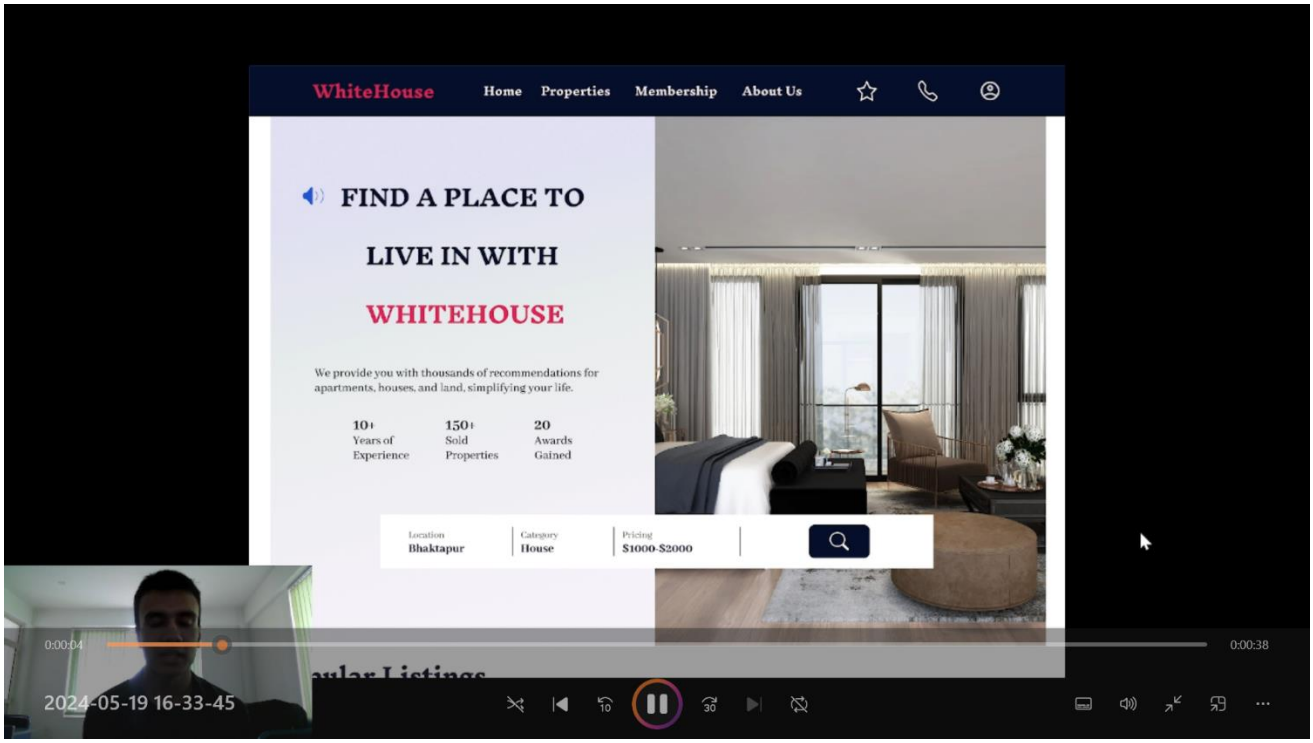


Figure 15: User testing-2

The screenshots above capture segments of the testing video conducted by three users. Feedback and areas for improvement were identified and discussed within the video, subsequently incorporated into the website design.

3.3 Usability Issues

Several usability issues were uncovered during testing could potentially impact the user experience they are:

- **Broken Links**

  Some pages had buttons or icons that were either unresponsive or only worked intermittently. Consequently, the broken links were repaired.

- **Inconsistent buttons**

  The buttons lacked consistency across the design's pages, varying in size and font, potentially confusing users and affecting their experience. Therefore, uniform sizes and fonts were implemented for the buttons throughout the interface.

- **Error and confirmation**

  A primary issue identified during user testing was the absence of a confirmation prompt during the payment process, which contravened Nielsen's usability Heuristics principle of error prevention. Upon clicking the button, even inadvertently, users were unable to reverse the purchase. Consequently, a confirmation popup was integrated post-user testing to mitigate errors.

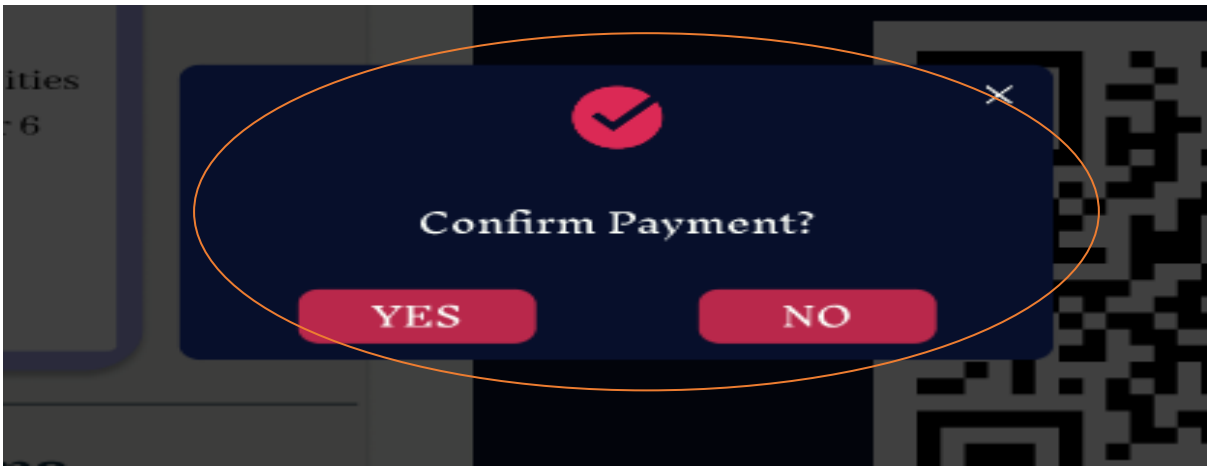


Figure 16: Confirmation

3.4 Accessibility Testing

Accessibility testing was conducted during user testing to determine if the WhiteHouse website is usable by individuals with disabilities, ensuring they can navigate it without encountering any obstacles. Several accessibility features my website are:

- **Text to Speech**

  The text-to-speech accessibility feature audibly reads the written content, which is advantageous for individuals with visual impairments.

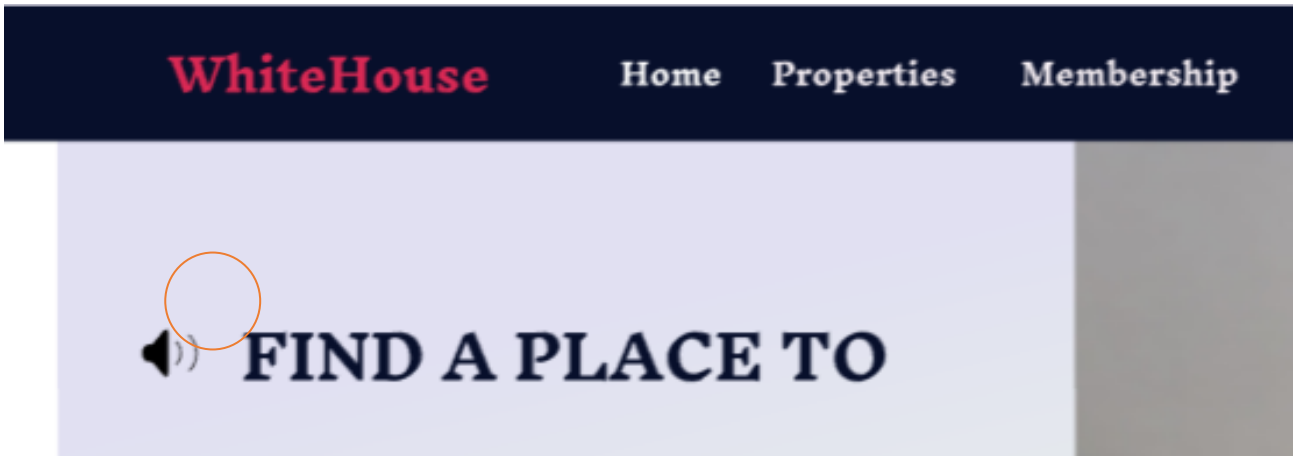


Figure 17: Accessibility Text-to-speech

- **Switch Language**

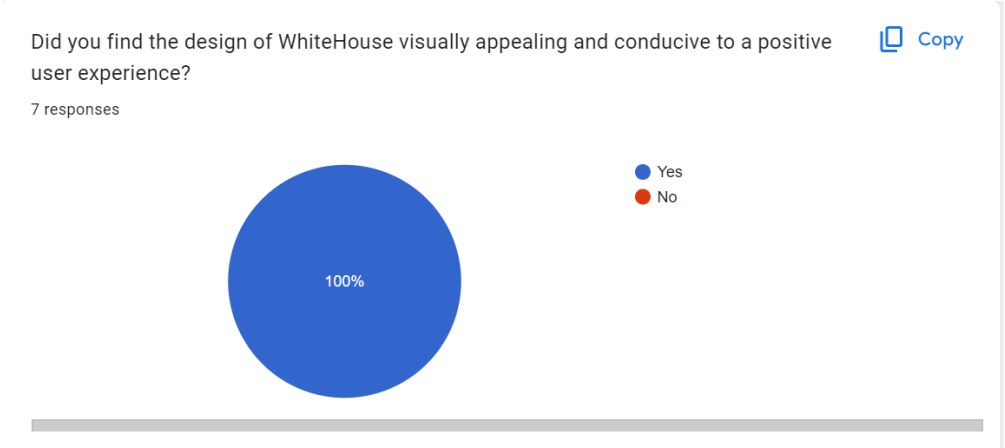
  The language switch feature enables user to convert text into a language they are comfortable with.

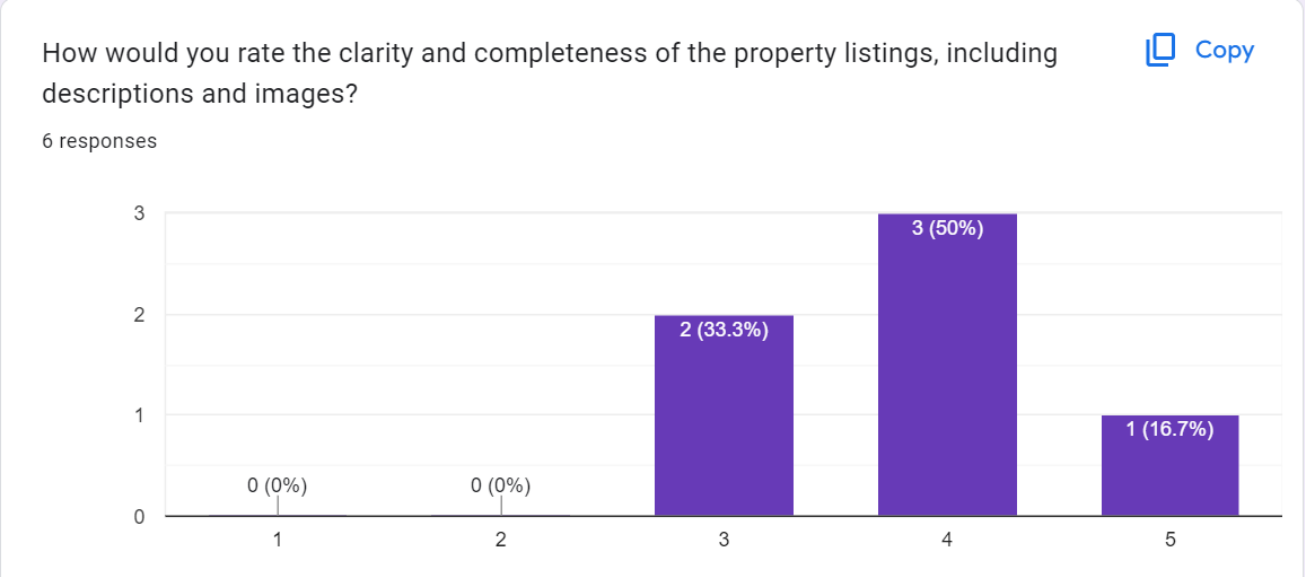
Figure 18: Accessibility-switch language

3.5 Usability and Accessibility Response Form

Two forms were given to the evaluators, usability testing form and accessibility testing form. The responses are as follows:

- **Usability**

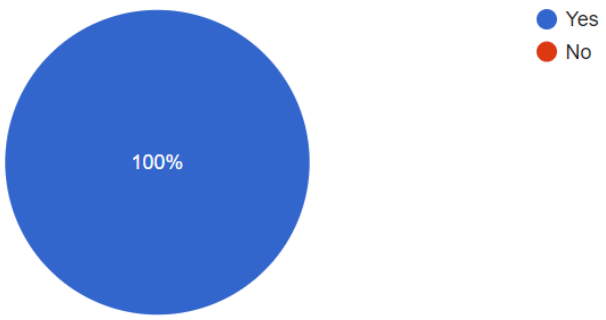




Were you able to easily access and use the interactive elements on property listings, such as virtual tours or maps?

Copy

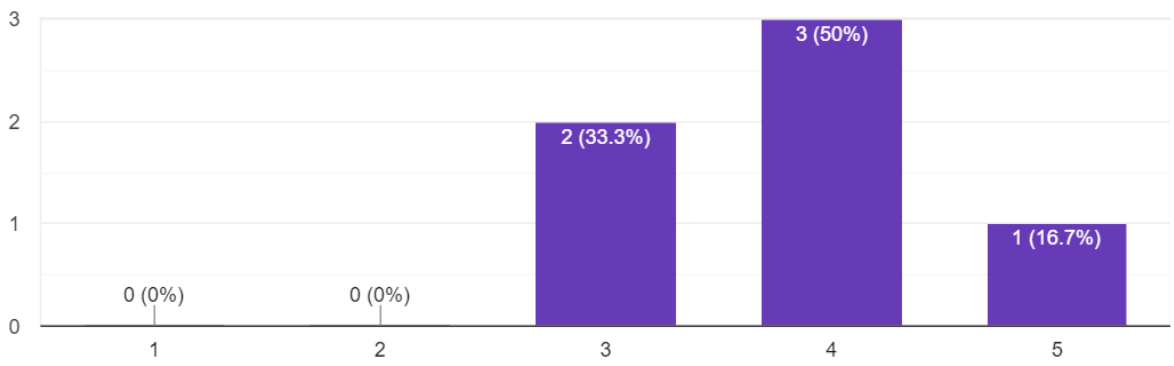
7 responses



How would you rate the clarity and completeness of the property listings, including descriptions and images?

Copy

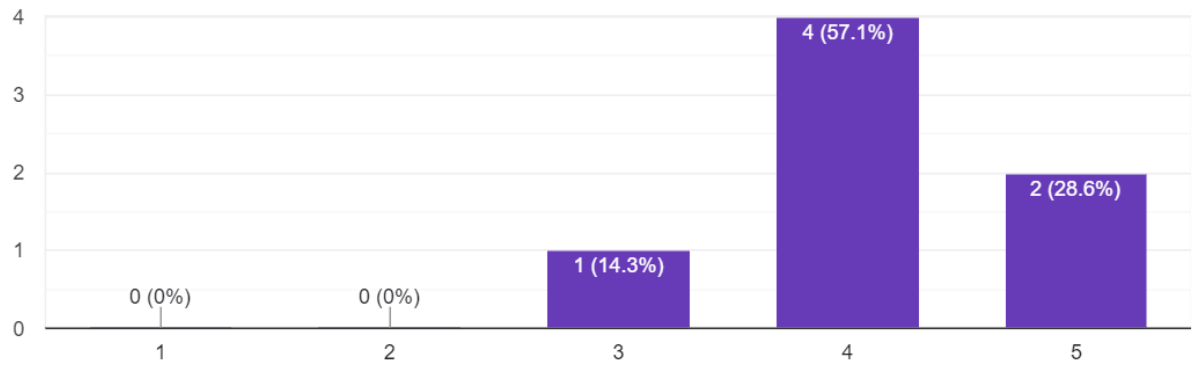
6 responses



Did you find the options available for advertising properties on WhiteHouse clear and easy to use?

Copy

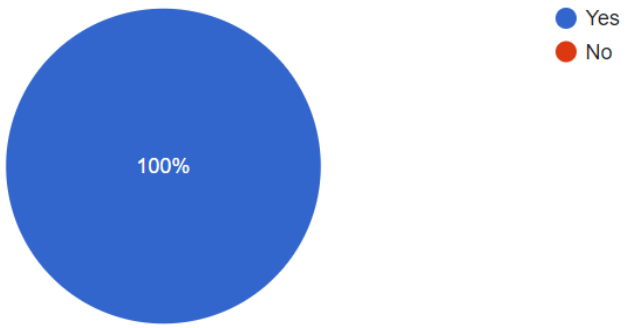
7 responses



Were you able to customize your property search effectively using the available filters and search options?

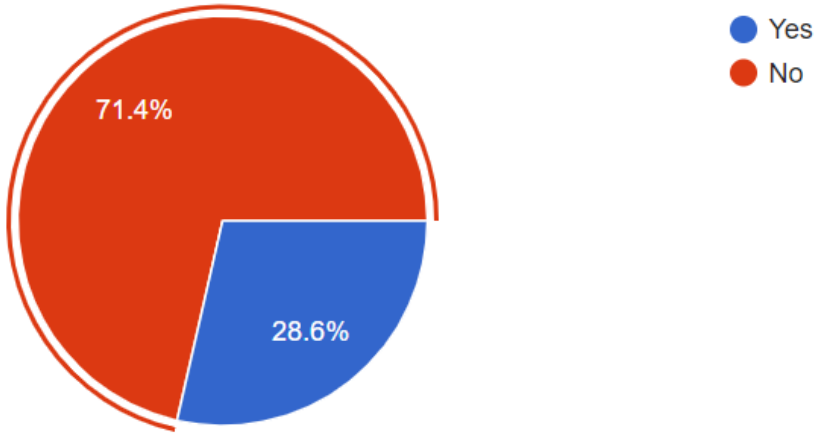
Copy

7 responses



Did you encounter any technical issues or bugs while using WhiteHouse?

7 responses



On a scale of 1 to 10, how satisfied are you with your overall experience using WhiteHouse for browsing and purchasing properties?

Copy

7 responses

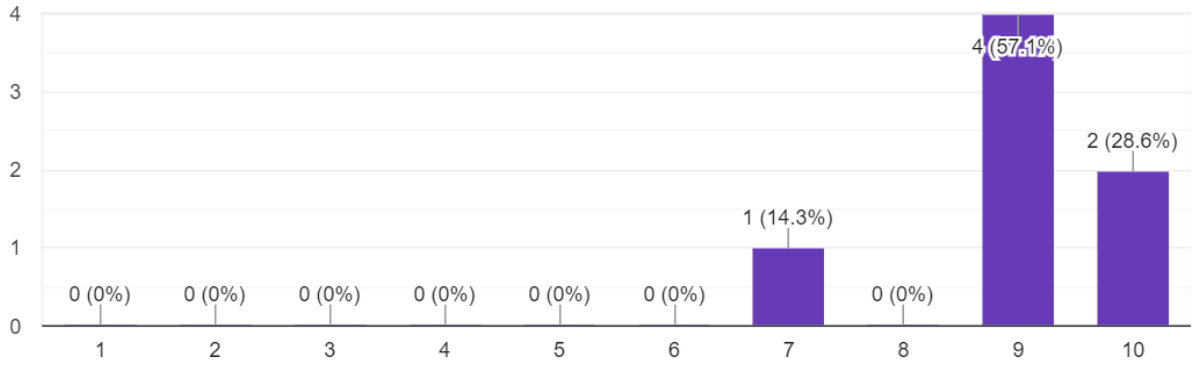


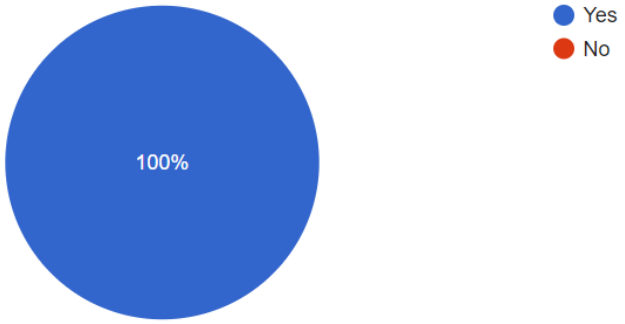
Figure 19: Usability Response

- **Accessibility**

Did you find the font sizes and color contrasts on WhiteHouse's website conducive to easy reading and navigation for all users?

Copy

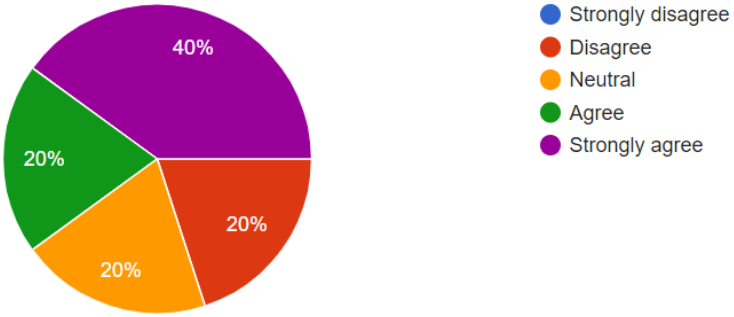
5 responses



Were the interactive elements, such as buttons and links, clearly labeled and easy to use for users with disabilities?

Copy

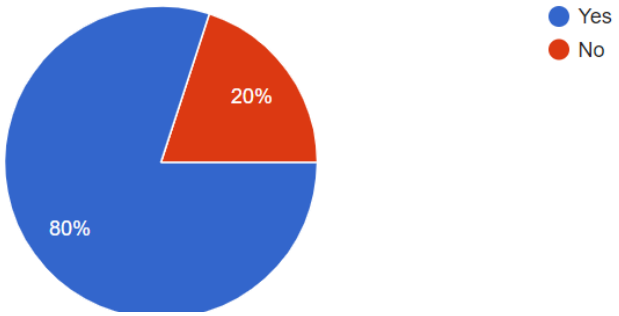
5 responses



Did you encounter any barriers or challenges related to accessibility while using WhiteHouse?

Copy

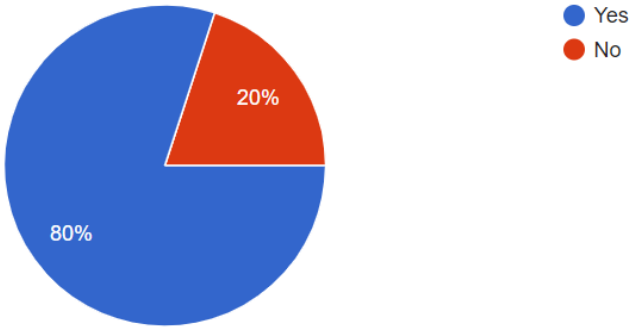
5 responses



Were the alt text descriptions for images on WhiteHouse sufficient for understanding their content and purpose?

Copy

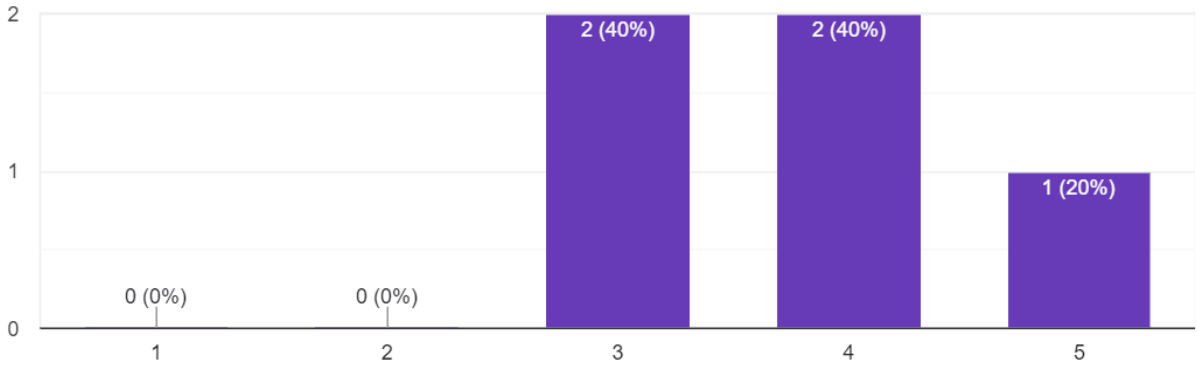
5 responses



How would you rate WhiteHouse's commitment to inclusivity and accessibility compared to other similar platforms or websites you've used?

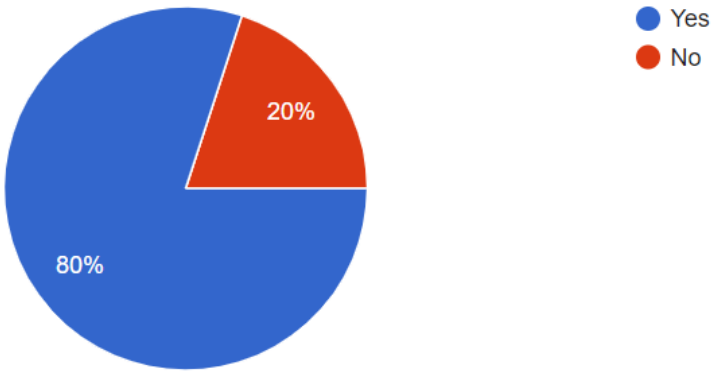
Copy

5 responses



How satisfied are you with the accessibility features provided by WhiteHouse, such as text-to-speech and language translation?

5 responses



Overall, how satisfied are you with WhiteHouse's accessibility efforts and its usability for users with disabilities?

Copy

5 responses

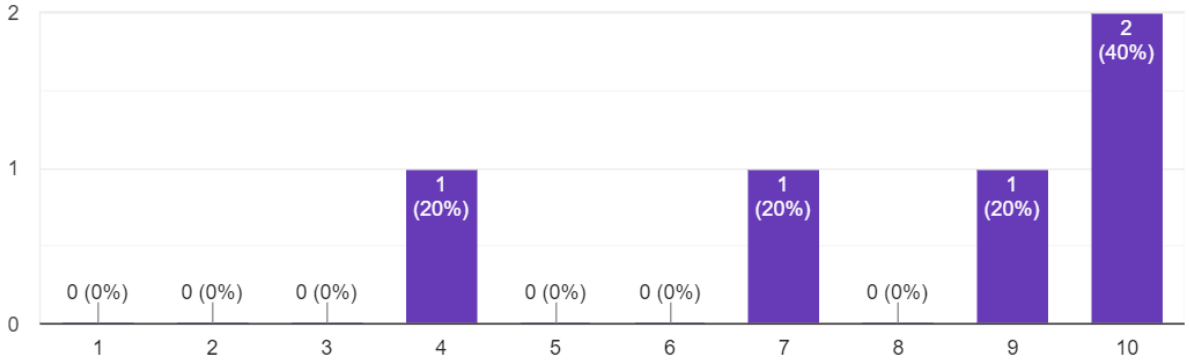


Figure 20: Accessibility Response

4. Conclusion

The WhiteHouse interface was crafted with an emphasis on simplicity and user-friendliness, ensuring accessibility for all users regardless of their abilities. Through brainstorming sessions, I generated numerous creative ideas and features for my interface. Developing user personas enabled me to recognize the demands of my client base, enabling facilities the creation of an initial design concept in the form of a wireframe. Using the wireframe and user flow, I developed an interactive prototype and conducted user testing. By adhering to these design processes, I was able to build an interface that exceeded my expectations.

During the user testing and design process, I found several accessibility issues and usability, including broken links, fickle buttons and fonts. After completing user testing, I focused on improving these functionalities. To further enhance the interface, I can introduce additional accessibility and usability features, such as conducting more user testing to collate responses to elevate the user experience and allowing users to compare properties in their favorites for better decision-making. Ultimately, by following the design process and addressing these issues, the WhiteHouse Website interface was created to both user-friendly and visually appealing, adhering to usability guidelines

5. Appendix

5.1 Testing Videos

- **Usability Testing**

  https://youtu.be/3gNYMWPA6Oc

- **Accessibility Testing**

https://youtu.be/HqWDymdkpDQ

https://youtu.be/_00JfTgOAOU

5.2 Form Responses

- **Usability Testing Form Responses**

https://docs.google.com/spreadsheets/d/1ahBRwOe9WqPNDx7zotx7MvHFx6rxG-h91K_jYe8pTy8/edit?usp=sharing

- **Accessibility Testing Form Response**

https://docs.google.com/spreadsheets/d/1ahBRwOe9WqPNDx7zotx7MvHFx6rxG-h91K_jYe8pTy8/edit?usp=sharing

5.3 Link to the Interactive Prototype

- **Website**

https://www.figma.com/design/oBvIC4QOOoH9yS0KtyYOBP/RealEstate?node-id=2-2&t=84TlWgRKkueeuluW-0

5.4 Wireframes

- **Home Page**

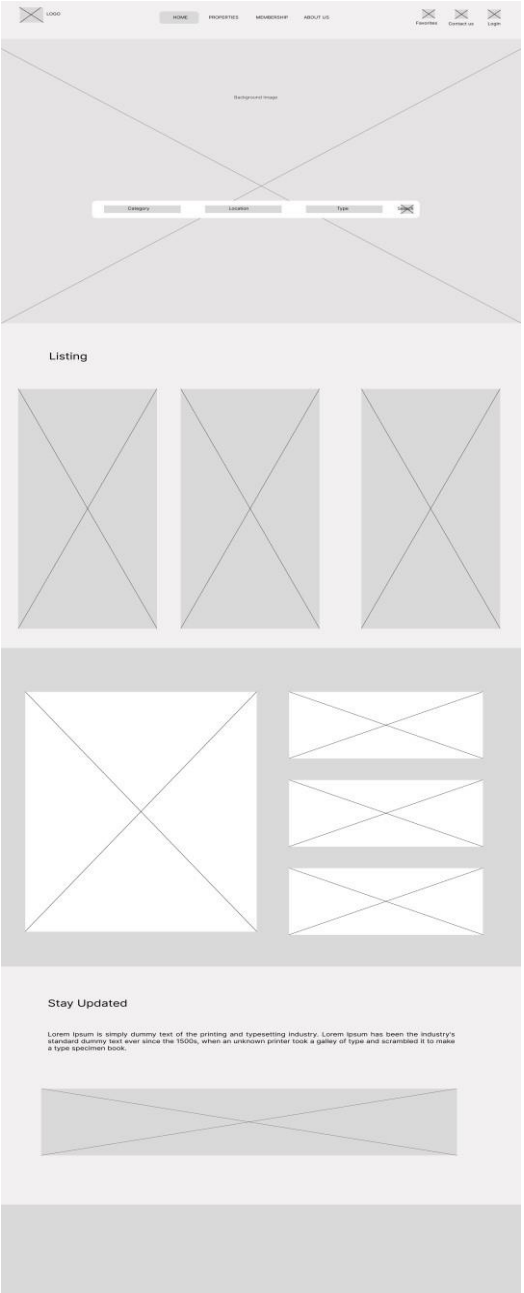


Figure 21: Wireframe-1

Property Page

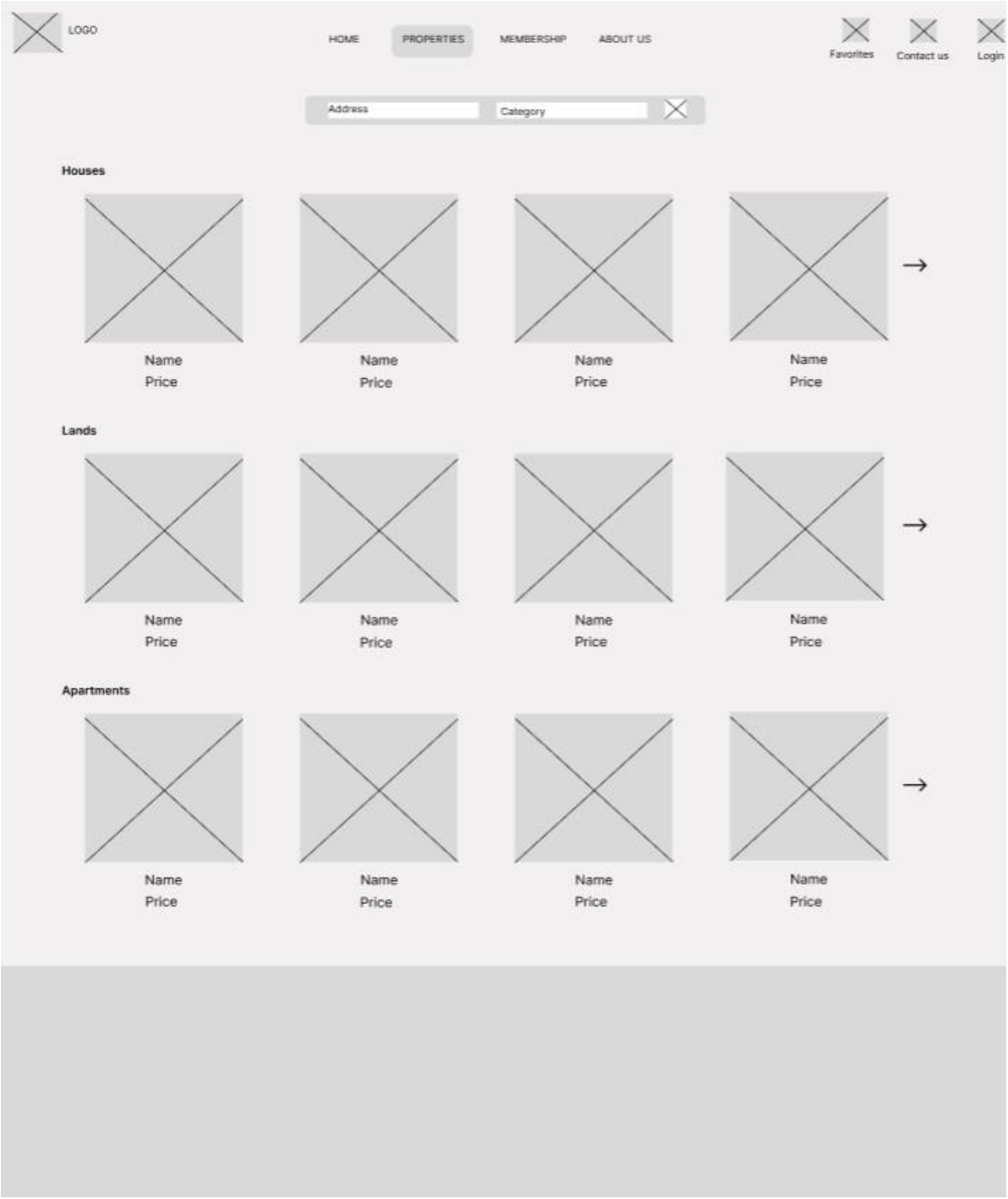


Figure 22: Wireframe-2

Login Page

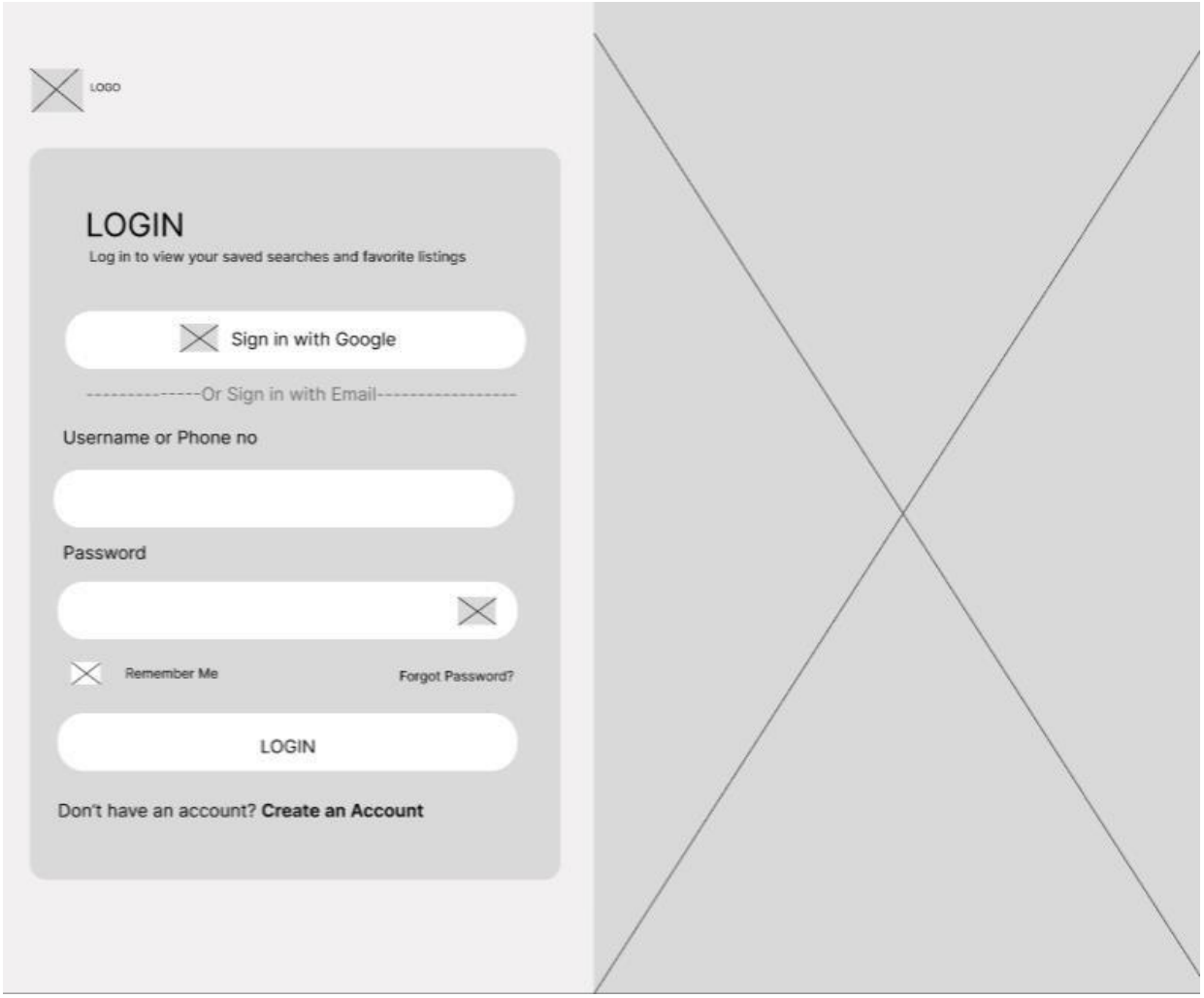


Figure 23: Wireframe-3

- **Membership Ad Page**

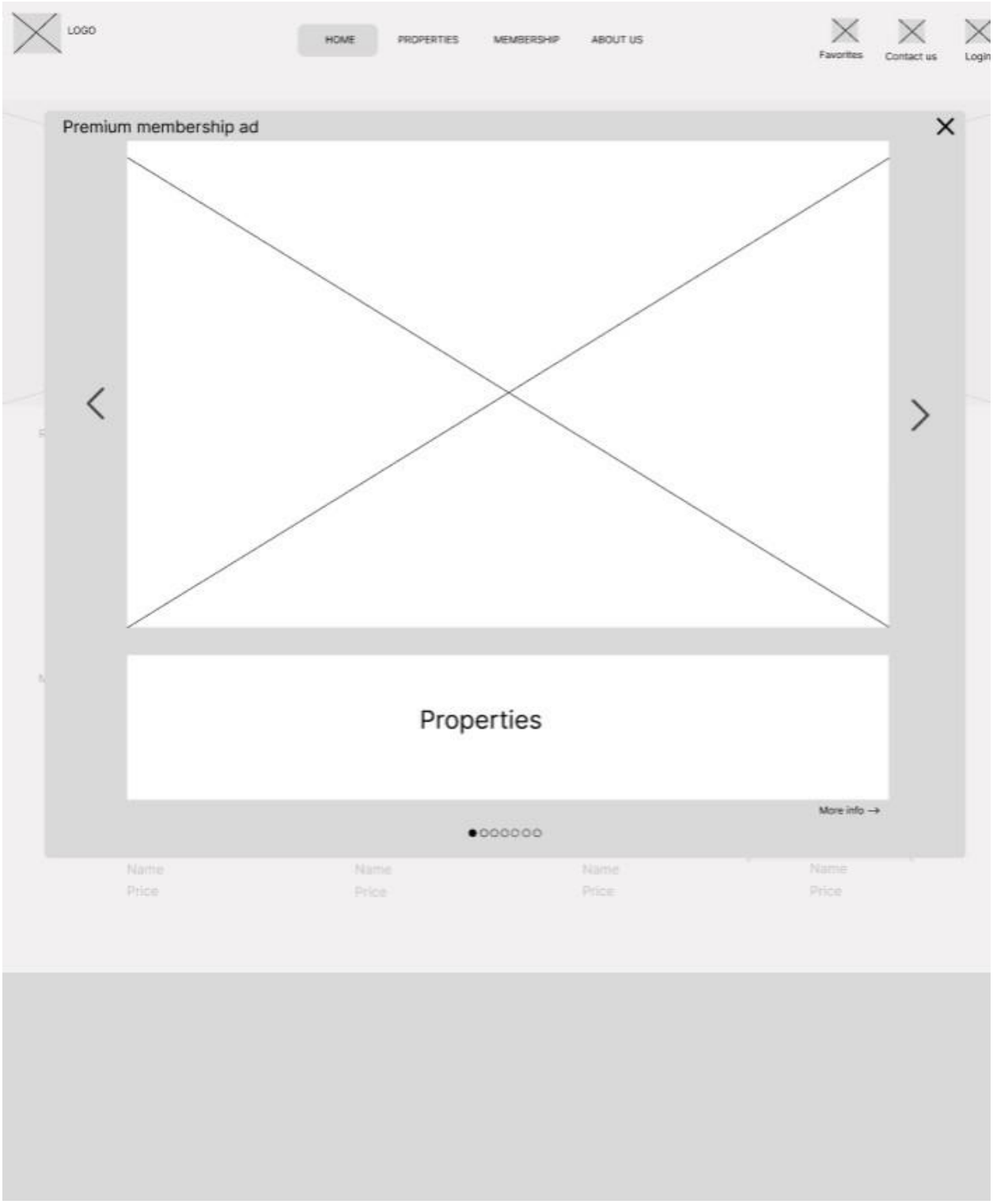


Figure 24: Wireframe-4

- **Membership Page**

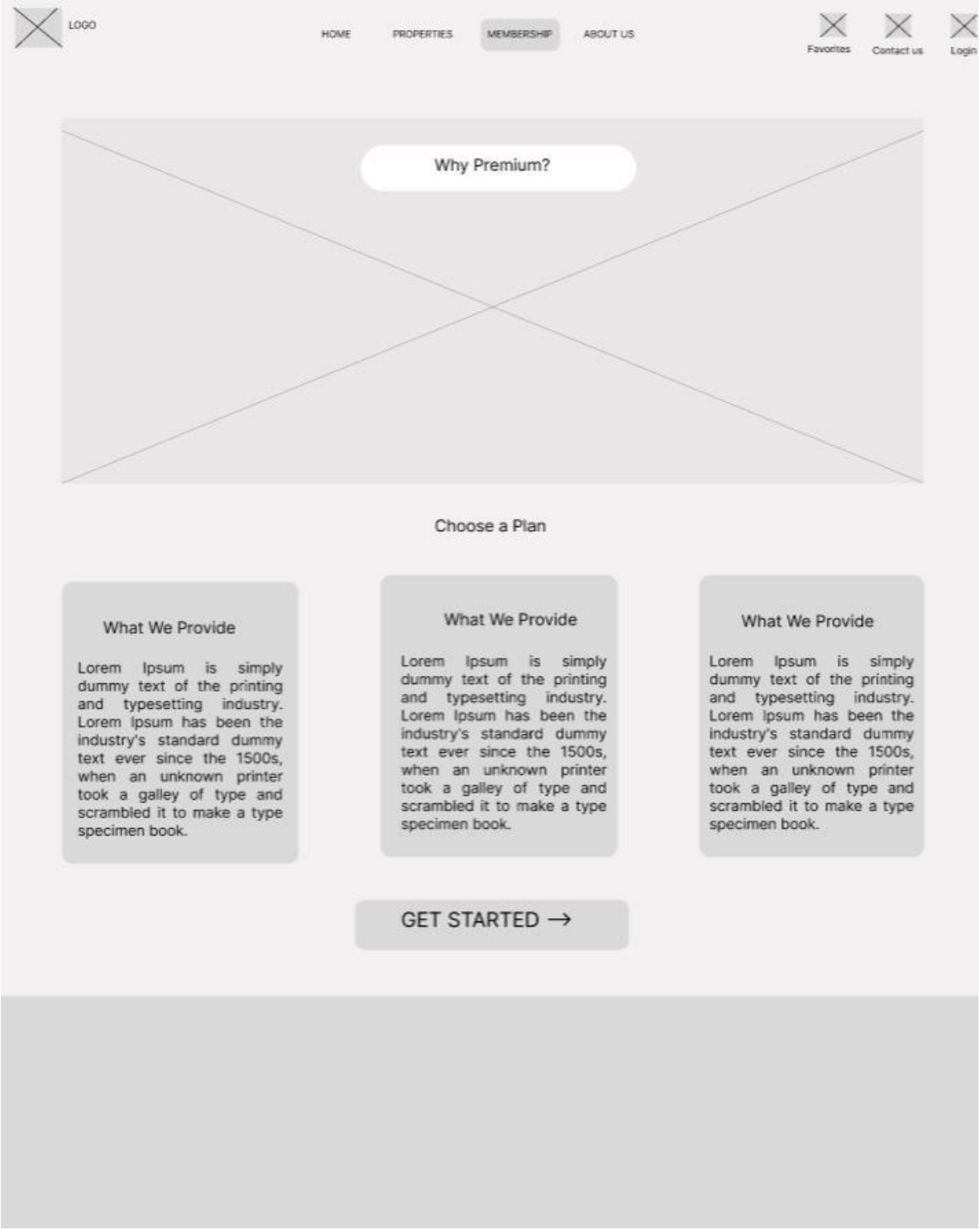


Figure 25: Wireframe-5

- **Payment Page**

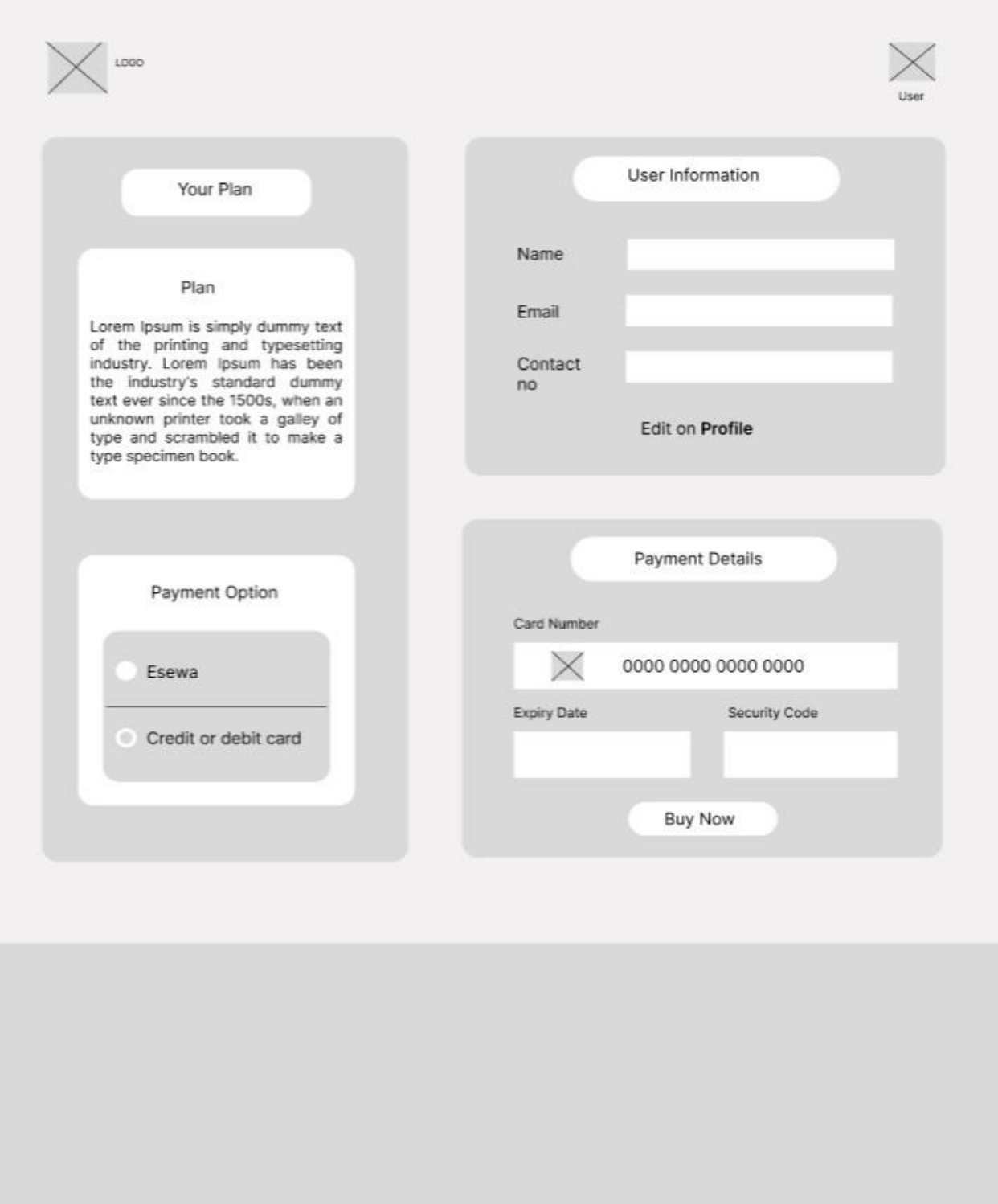


Figure 26: Wireframe-6

- **Schedule Meeting Page**

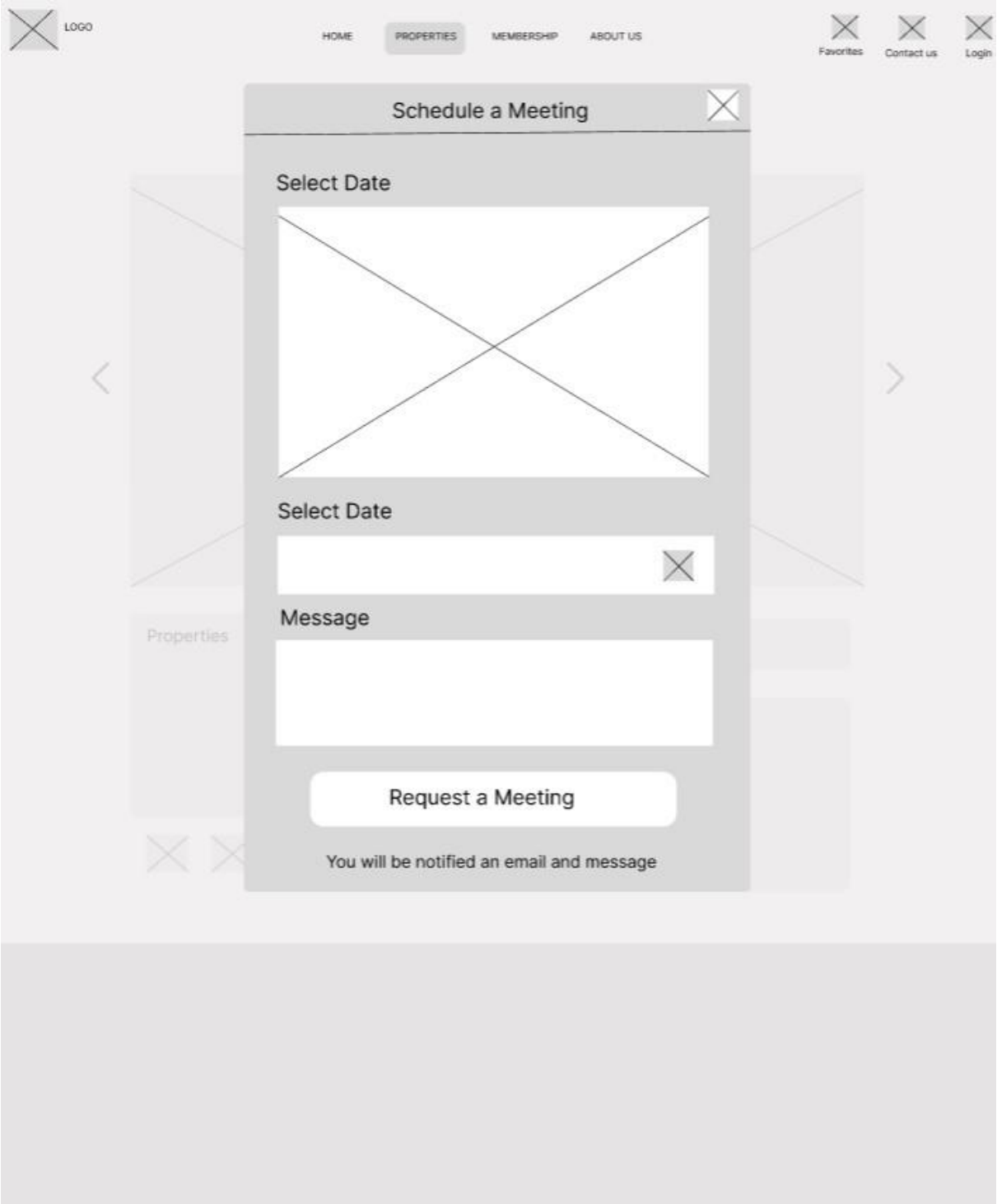


Figure 27: Wireframe-7

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About Us Page

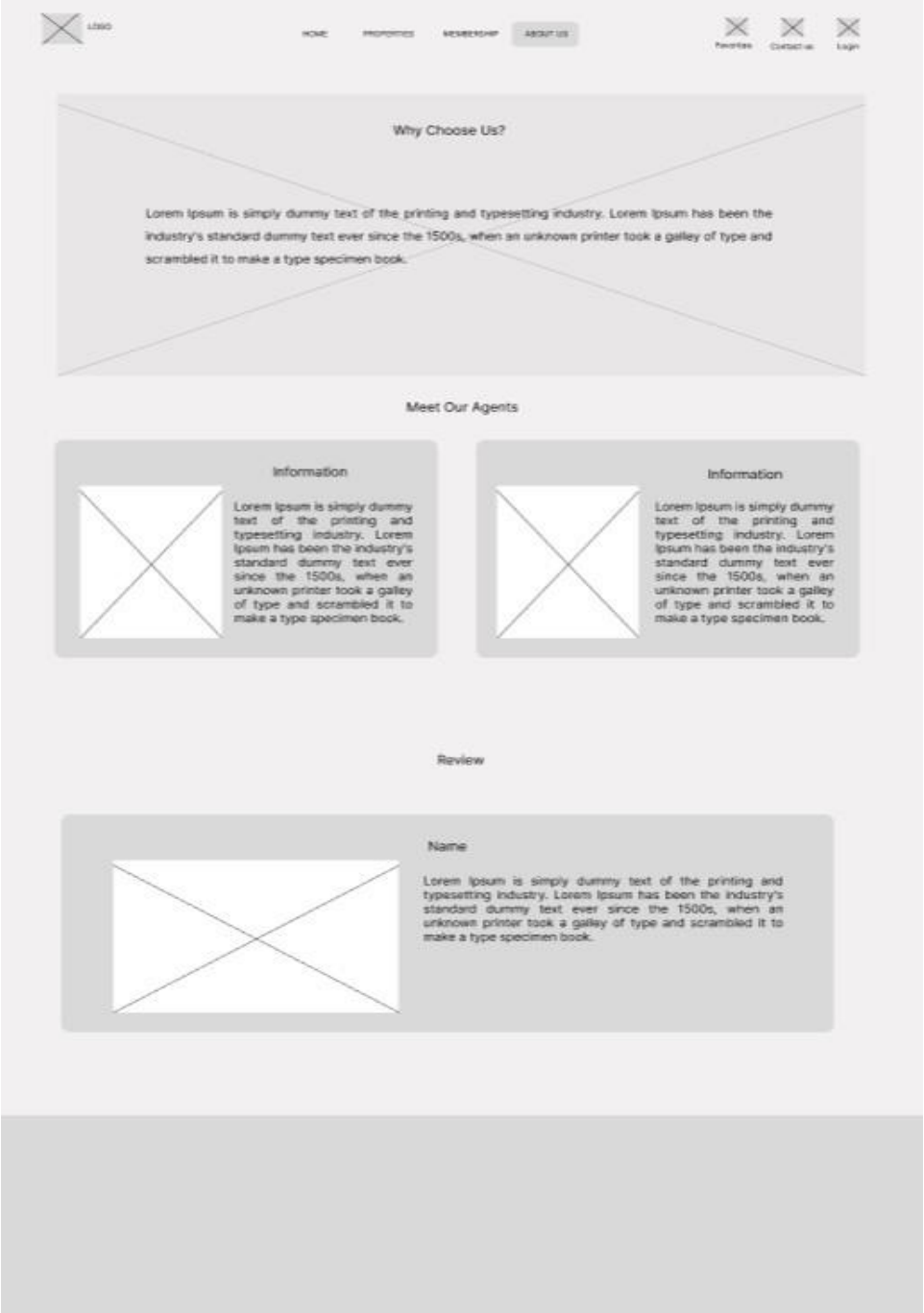


Figure 28: Wireframe-8

- **Favorite Page**

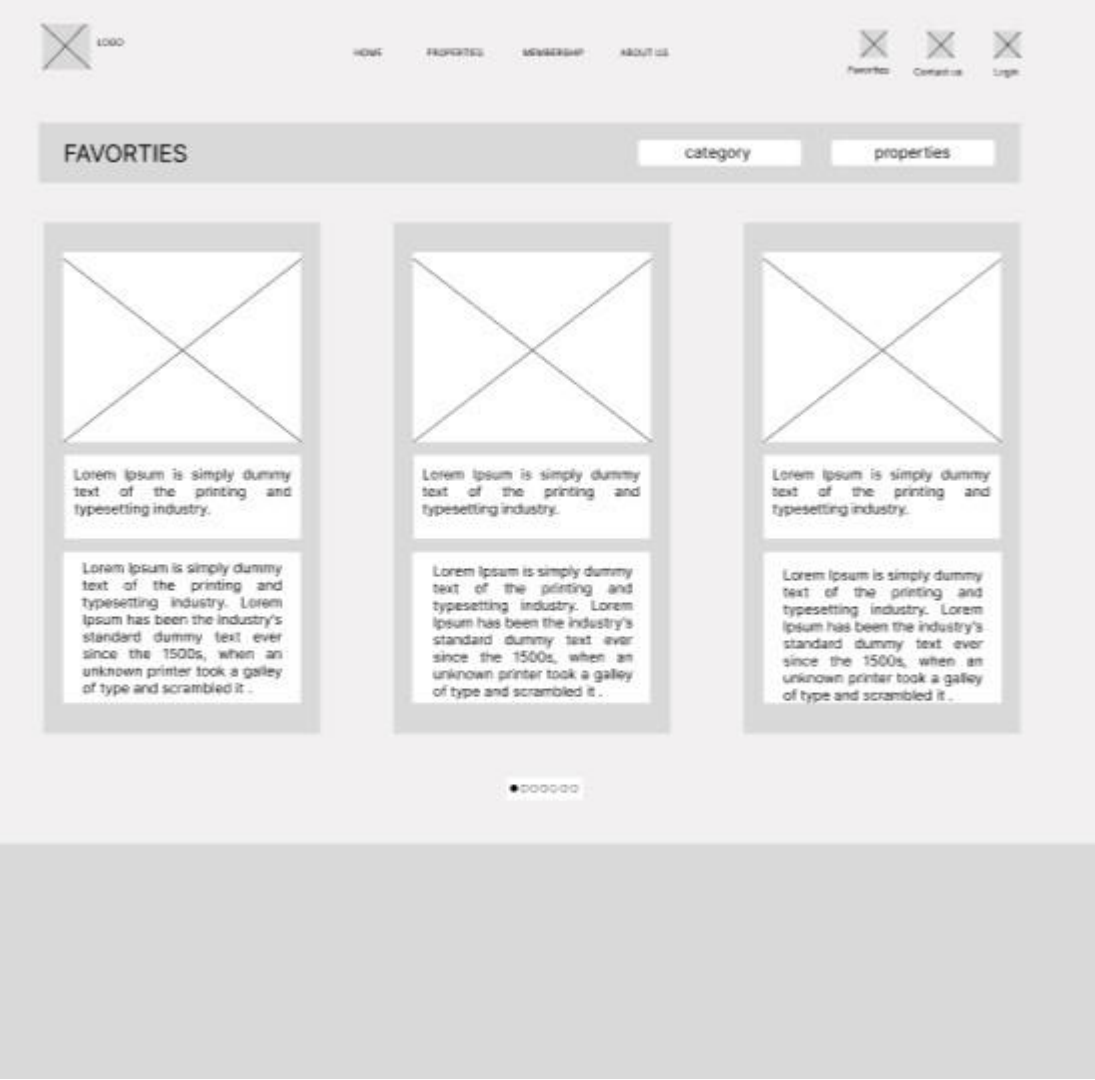


Figure 29: Wireframe-9

- **Payment Option Page**

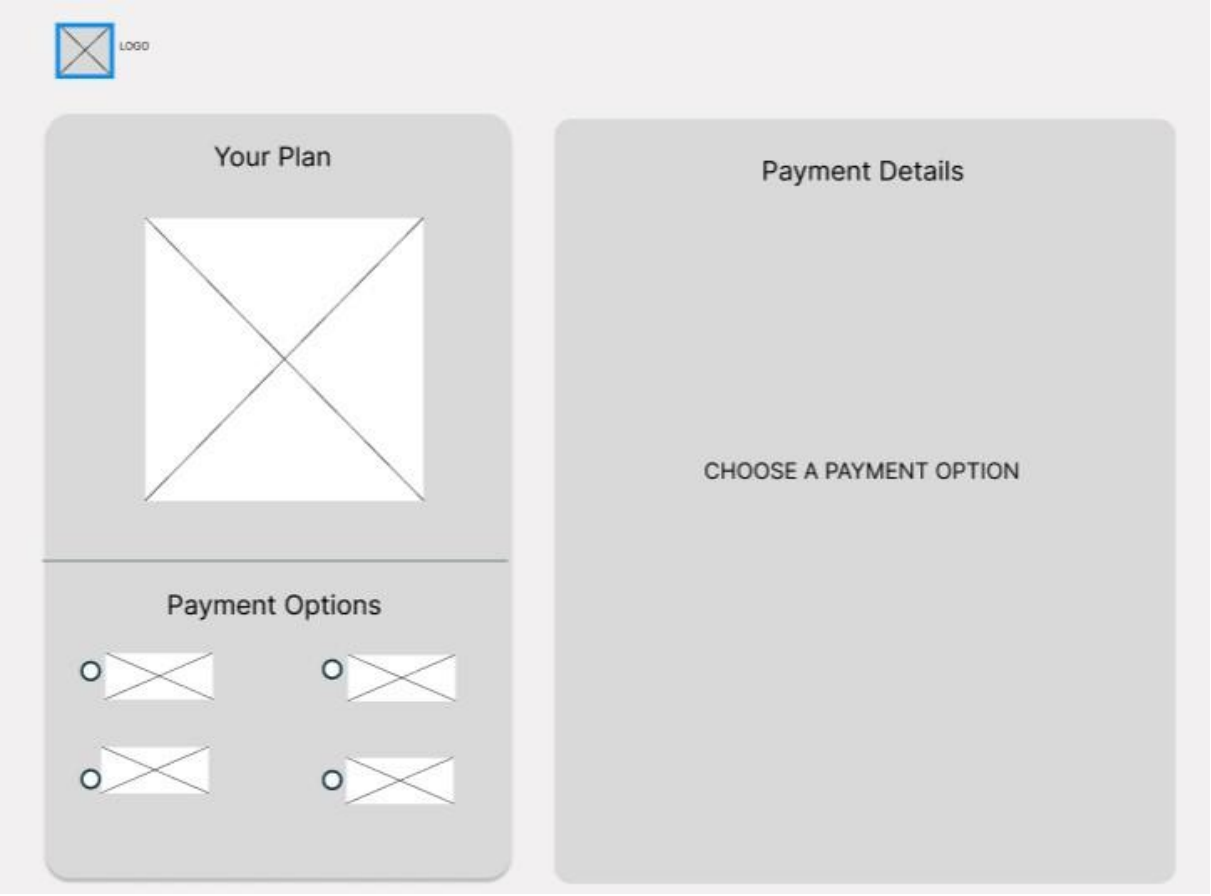
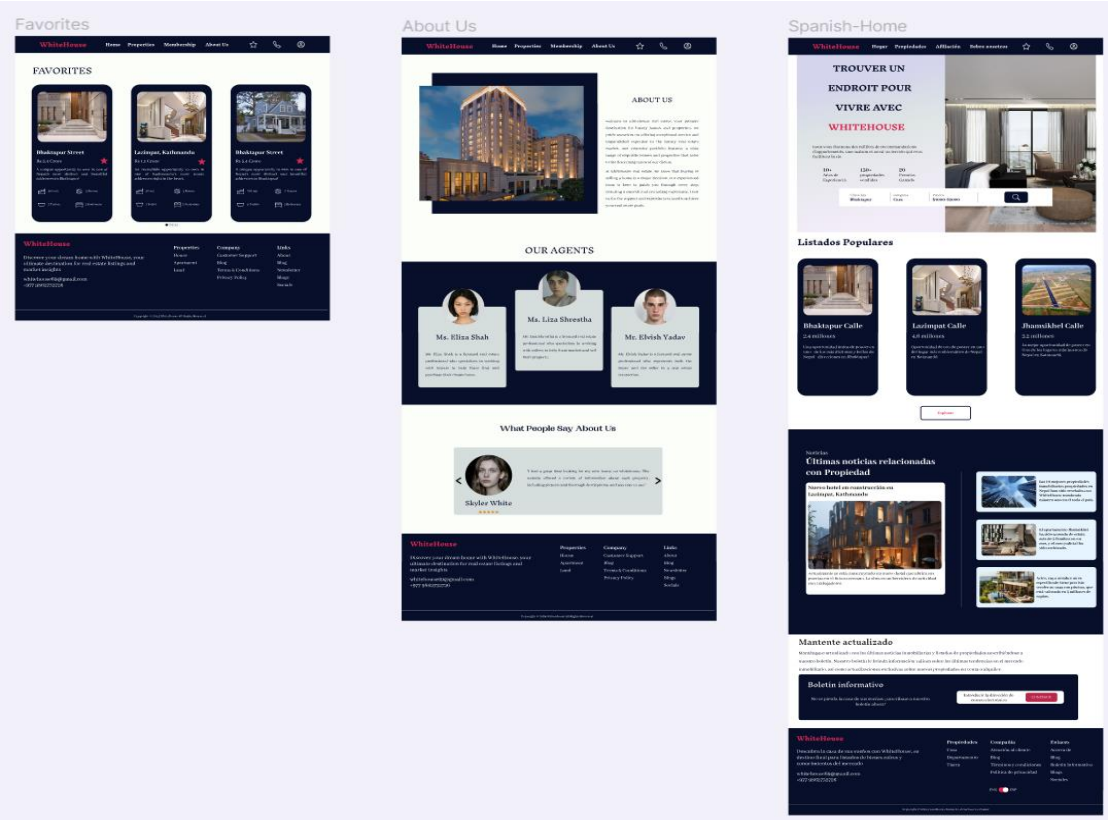
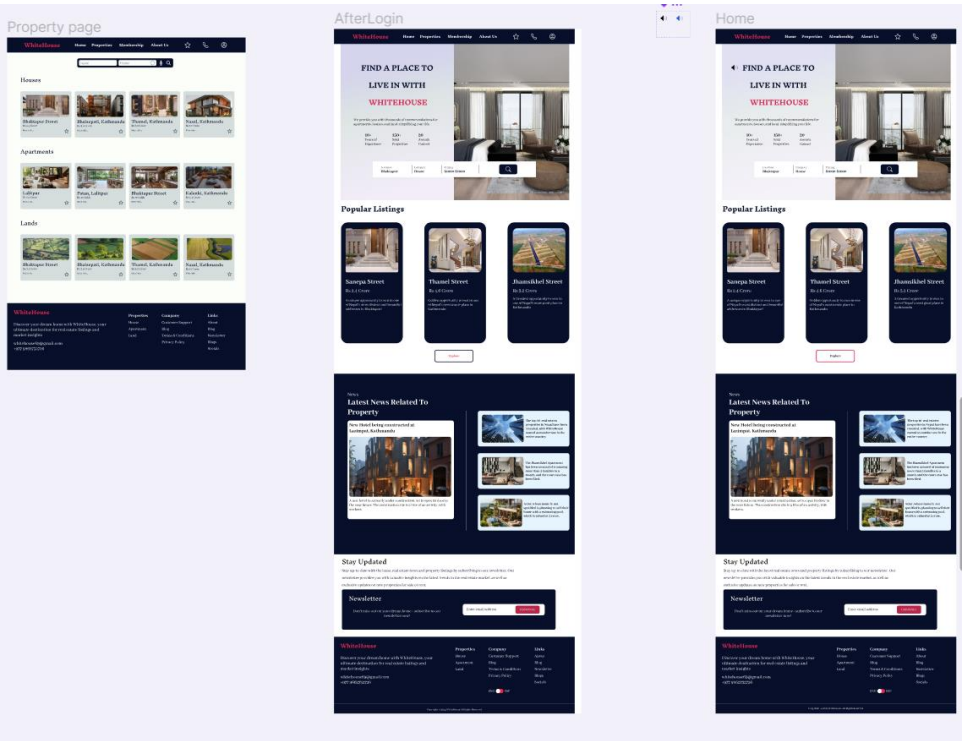


Figure 30: Wireframe-10

5.5 Design


Favorites
FAVORITES
About Us
ABOUT US
OUR AGENTS
Ms. Eliza Shah
Ms. Liza Shrestha
Mr. Elvish Yadav
What People Say About Us
Skyler White
Spanish-Home
TROUVER UN
ENDROIT POUR
VIVRE AVEC
WHITEHOUSE
Listados Populares
Bhaktapur Calle
Lazimpat Calle
Jhamsikhel Calle
Últimas noticias relacionadas con Propiedad
Mantente actualizado
Boletín informativo


Property page
Houses
Apartments
Lands
AfterLogin
FIND A PLACE TO
LIVE IN WITH
WHITEHOUSE
Popular Listings
Latest News Related To Property
Stay Updated
Newsletter
Home
FIND A PLACE TO
LIVE IN WITH
WHITEHOUSE
Popular Listings
Latest News Related To Property
Stay Updated
Newsletter

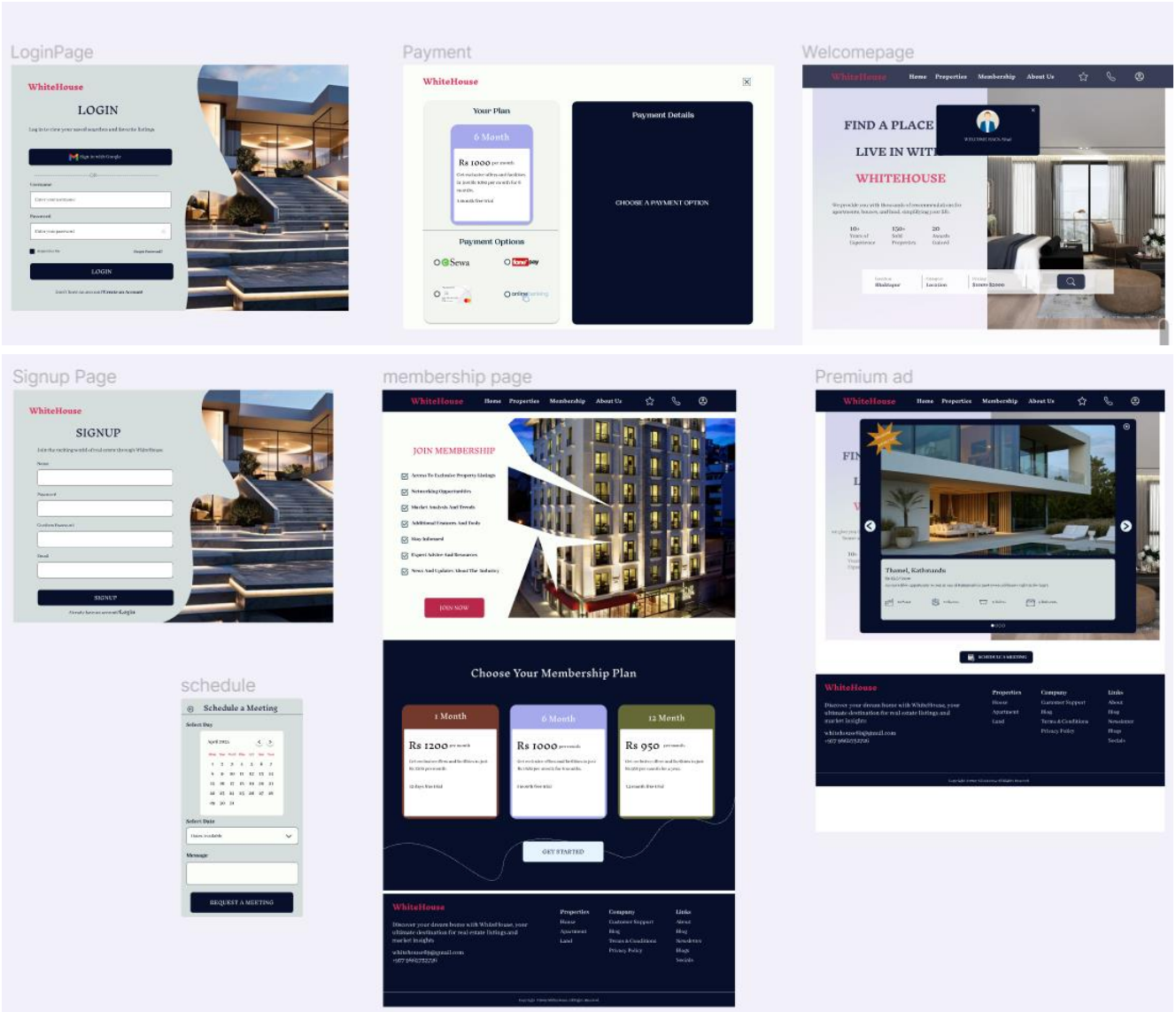


Figure 31: Design